

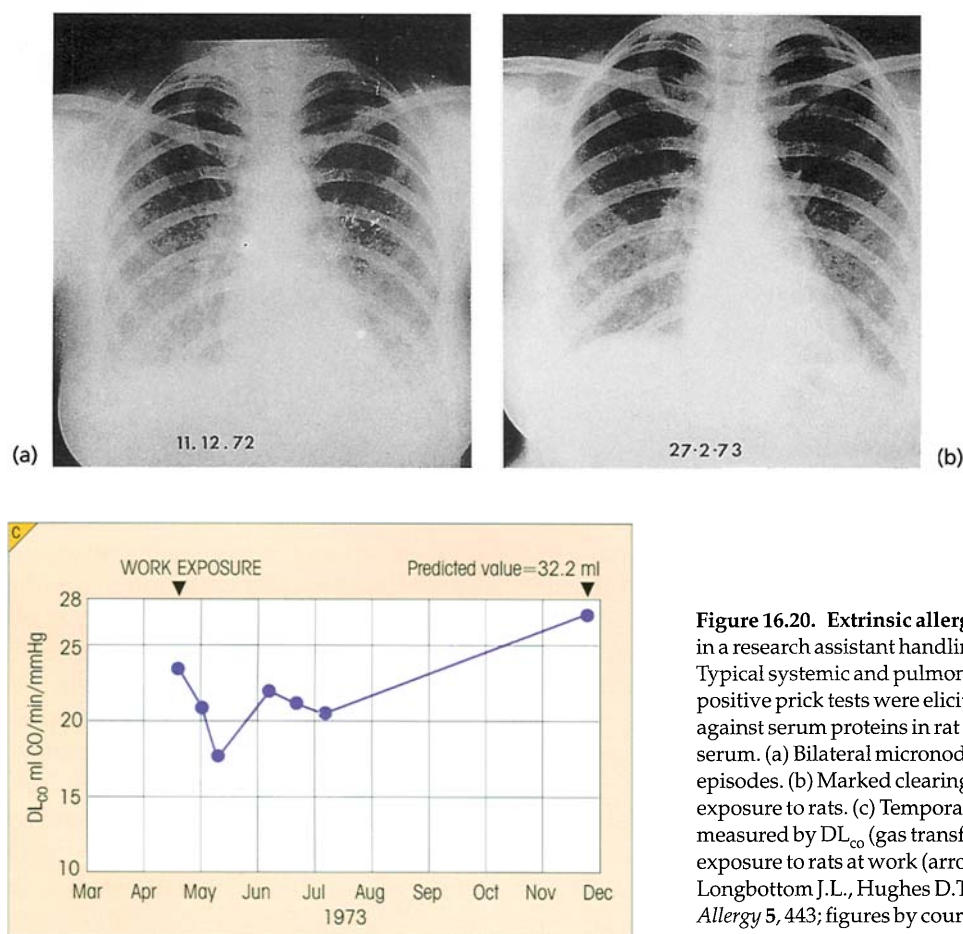


Figure 16.20. Extrinsic allergic alveolitis due to rat serum proteins in a research assistant handling rats (type III hypersensitivity). Typical systemic and pulmonary reactions on inhalation and positive prick tests were elicited by rat serum proteins; precipitins against serum proteins in rat urine were present in the patient's serum. (a) Bilateral micronodular shadowing during acute episodes. (b) Marked clearing within 11 days after cessation of exposure to rats. (c) Temporary fall in pulmonary gas exchange measured by DL_{co} (gas transfer, single breath) following a 3-day exposure to rats at work (arrowed). (From Carroll K.B., Pepys J., Longbottom J.L., Hughes D.T.D. & Benson H.G. (1975) *Clinical Allergy* 5, 443; figures by courtesy of Professor J. Pepys.)

quaintly named cases of **extrinsic allergic alveolitis** resulting from continual inhalation of organic particles, e.g. cheese washer's disease (*Penicillium casei* spores), furrier's lung (fox fur proteins) and maple bark stripper's disease (spores of *Cryptostroma*). Evidence that an immediate anaphylactic type I response may sometimes be of importance for the initiation of an Arthus reaction comes from the study of patients with allergic bronchopulmonary aspergillosis who have high levels of IgE and precipitating IgG antibodies to *Aspergillus* species.

Reactions to internal antigens

Type III reactions are often provoked by the local release of antigen from infectious organisms within the body; for example, living filarial worms, such as *Wuchereria bancrofti*, are relatively harmless, but the dead parasite found in lymphatic vessels initiates an inflammatory reaction thought to be responsible for the obstruction of lymph flow and the ensuing, rather monstrous, elephantiasis. Chemotherapy may cause an abrupt release of microbial antigens in individuals with high antibody levels, producing quite dramatic



Figure 16.21. Erythema nodosum leprosum, forearm. The patient has lepromatous leprosy with superimposed erythema nodosum leprosum. These acutely inflamed nodules were extremely tender and the patient was pyrexial. (Photograph kindly provided by Dr G. Levene.)

immune complex-mediated reactions, such as **erythema nodosum leprosum** in the skin of dapsone-treated lepromatous leprosy patients (figure 16.21) and the Jarisch–Herxheimer reaction in syphilitics on penicillin.

An interesting variant of the Arthus reaction is seen in rheumatoid arthritis where complexes are formed locally in the joint due to the production of self-associating IgG anti-IgG by synovial plasma cells (cf. p. 429).

It has also been recognized that complexes could be generated at a local site by a quite different mechanism involving nonspecific adherence of an antigen to tissue structures followed by the binding of soluble antibody—in other words, the antigen becomes fixed in the tissue *before* not *after* combining with antibody. Although it is not clear to what extent this mechanism operates in patients with immune complex disease, let us describe the experimental observation on which it is based. After injection with bacterial endotoxin, mice release DNA into their circulation which binds specifically to the collagen in the basement membrane of the glomerular capillaries; the endotoxin also polyclonally activates B-cells making anti-DNA which gives rise to antigen–antibody complexes in the kidney.

Disease resulting from circulating complexes

Immune complex glomerulonephritis

The deposition of complexes is a dynamic affair and

long-lasting disease is only seen when the antigen is persistent, as in chronic infections and autoimmune diseases. Experimentally, Dixon produced chronic glomerular lesions by repeated administration of foreign proteins to rabbits. Not all animals showed the lesion, and perhaps only those genetically capable of producing low affinity antibody or antibodies to a restricted number of determinants formed soluble complexes in the right size range. The **smallest complexes reach the epithelial side**, but progressively **larger complexes are retained in or on the endothelial side of the glomerular basement membrane** (figure 16.22). They build up as ‘lumpy’ granules staining for antigen, immunoglobulin and complement (C3) by immunofluorescence (figure 16.17b), and appear as large amorphous masses in the electron microscope (cf. figure 20.5). The inflammatory process damages the basement membrane through engagement of the complexes with effector cells bearing Fc γ receptors, as revealed by the absence of glomerulonephritis despite immune complex deposition in the kidneys of Fc γ R-deficient New Zealand (B $\times$ W) F1 hybrids (a murine model of human systemic lupus erythematosus, SLE; p. 403). Proteinuria results from the leakage of serum proteins through the damaged membrane and serum albumin, being small, appears in the urine even with

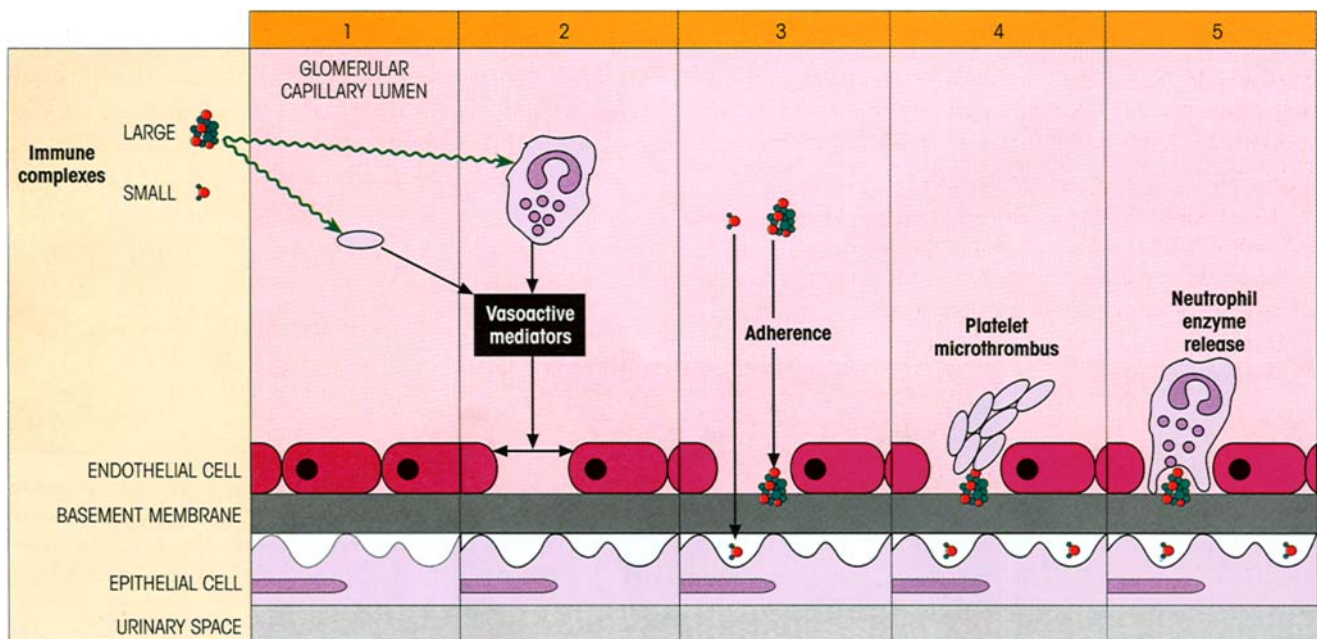


Figure 16.22. Deposition of immune complexes in the kidney glomerulus. (1) Complexes induce release of vasoactive mediators from basophils and platelets which cause (2) separation of endothelial cells. (3) Attachment of larger complexes to exposed basement membrane, with smaller complexes passing through to the epithelial side. (4) Complexes induce platelet aggregation. (5) Chemotactically

attracted neutrophils release granule contents in ‘frustrated phagocytosis’ to damage basement membrane and cause leakage of serum proteins. Complex deposition is favored in the glomerular capillary because it is a major filtration site and has a high hydrodynamic pressure. Deposition is greatly reduced in animals depleted of platelets or treated with vasoactive amine antagonists.

minor degrees of glomerular injury (figure 16.23, lane 3).

Many cases of glomerulonephritis are associated with circulating complexes, and biopsies give a fluorescent staining pattern similar to that of figure 16.17b, which depicts DNA/anti-DNA/complement deposits in the kidney of a patient with SLE (cf. p. 427). Well known is the disease which can follow infection with certain strains of so-called 'nephritogenic' streptococci and the nephrotic syndrome of Nigerian children associated with quartan malaria, where com-

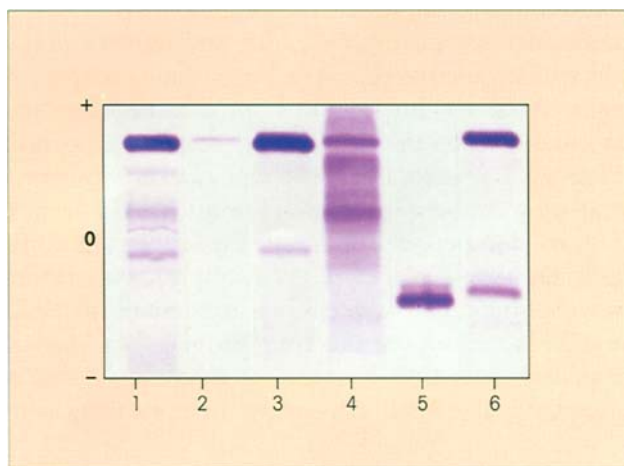


Figure 16.23. Proteinuria demonstrated by electrophoresis. Lane 1: Normal serum as reference. The major band nearest to the cathode is albumin. Lane 2: Normal urine showing a trace of albumin. Lane 3: Glomerular proteinuria showing a major albumin component. Lane 4: Proteinuria resulting from tubular damage with a totally different electrophoretic pattern. Lane 5: Bence-Jones proteinuria representing excreted paraprotein light chains (cf. p. 385). Lane 6: Bence-Jones proteinuria with a trace of the intact paraprotein. Some of the samples have been concentrated. (Electropherograms kindly supplied by T. Heys.)

plexes with antigens of the infecting organism have been implicated. Immune complex nephritis can arise in the course of chronic viral infections; for example, mice infected with lymphocytic choriomeningitis virus develop a glomerulonephritis associated with circulating complexes of virus and antibody. This may well represent a model for many cases of glomerulonephritis in humans.

Deposition of immune complexes at other sites

The choroid plexus, being a major filtration site, is also favored for immune complex deposition and this could account for the frequency of central nervous disorders in SLE. Neurologically affected patients tend to have depressed C4 in the cerebrospinal fluid (CSF) and, at post-mortem, SLE patients with neurologic disturbances and high titer anti-DNA were shown to have scattered deposits of immunoglobulin and DNA in the choroid plexus. Subacute sclerosing panencephalitis is associated with a high CSF to serum ratio of measles antibody, and deposits containing Ig and measles Ag may be found in neural tissue.

Vasculitic skin rashes are also characteristic of systemic and discoid lupus erythematosus (figure 16.24a and b), and biopsies of the lesions reveal amorphous deposits of Ig and C3 at the basement membrane of the dermal-epidermal junction (cf. figure 20.6).

Another example of immune complex hypersensitivity is the hemorrhagic shock syndrome found with some frequency in South-East Asia during a second infection with a dengue virus. There are four types of virus, and antibodies to one type produced during a first infection may not neutralize a second strain but rather facilitate its entry into, and replication within, human monocytes by attachment of the complex to



Figure 16.24. Vasculitic skin rashes due to immune complex deposition. (a) Facial appearance in systemic lupus erythematosus (SLE). Lesions of recent onset are symmetrical, red and edematous. They are often most pronounced on the areas of the face which receive most light exposure, i.e. the upper cheeks and bridge of the nose, and the prominences of the forehead. (b) Vasculitic lesions in SLE. Small purpuric macules are seen.

Fc receptors. The enhanced production of virus leads to immune complex formation and a massive intravascular activation of the classical complement pathway. In some instances drugs such as penicillin become antigenic after conjugation with body proteins and form complexes which mediate hypersensitivity reactions.

It should be said that persistence of circulating complexes does not invariably lead to type III hypersensitivity (e.g. in many cancer patients and in individuals with idiotypic–anti-idiotypic reactions). Perhaps in these cases the complexes lack the ability to initiate the changes required for complex deposition, but some hold the view that complexes detected in the serum may sometimes be artifacts released from their *in vivo* attachment to the erythrocyte CR1 receptors by the action of factor I during processing of the blood.

Treatment

The avoidance of exogenous inhaled antigens inducing type III reactions is obvious. Elimination of microorganisms associated with immune complex disease by chemotherapy may provoke a further reaction due to copious release of antigen. Suppression of the accessory factors thought to be necessary for the deposition of complexes would seem logical. Disodium cromoglycate, heparin and salicylates are often used, the latter being an effective platelet stabilizer as well as a potent anti-inflammatory agent. Corticosteroids are particularly powerful inhibitors of inflammation and are immunosuppressive. In many cases, particularly those involving autoimmunity, conventional immunosuppressive agents may be justified. Where type III hypersensitivity is thought to arise

from an inadequate immune response, the more aggressive approach of immunopotentialization to boost avidity is being advocated, but that is a path that will be trodden gently.

CELL-MEDIATED (DELAYED-TYPE) HYPERSENSITIVITY (TYPE IV)

This form of hypersensitivity is encountered in many allergic reactions to bacteria, viruses and fungi, in the contact dermatitis resulting from sensitization to certain simple chemicals and in the rejection of transplanted tissues. Perhaps the best known example is the **Mantoux reaction** obtained by injection of tuberculin into the skin of an individual in whom previous infection with the mycobacterium had induced a state of cell-mediated immunity (CMI). The reaction is characterized by erythema and induration (figure 16.25a) which appears only after several hours (hence the term ‘delayed’) and reaches a maximum at 24–48 hours, thereafter subsiding. Histologically, the earliest phase of the reaction is seen as a perivascular cuffing with mononuclear cells followed by a more extensive exudation of mono- and polymorphonuclear cells. The latter soon migrate out of the lesion leaving behind a predominantly mononuclear cell infiltrate consisting of lymphocytes and cells of the monocyte–macrophage series (figure 16.25b). This contrasts with the essentially ‘neutrophil’ character of the Arthus reaction (figure 16.19b).

Comparable reactions to soluble proteins are obtained when sensitization is induced by incorporation of the antigen into complete Freund’s adjuvant (see p. 192). In some, but not all cases, if animals are primed with antigen alone or in incomplete Freund’s adjuvant



Figure 16.25. Cell-mediated (type IV) hypersensitivity reactions. (a) Mantoux test showing cell-mediated hypersensitivity reaction to tuberculin, characterized by induration and erythema. (b) Chronic type IV inflammatory lesion in tuberculous lung showing caseous necrosis (CN), epithelioid cells (E), giant cells (G) and mononuclear inflammatory cells (M). (c) Type IV contact hypersensitivity reaction

to nickel caused by the clasp of a necklace. ((a) Kindly provided by Professor J. Brostoff and (b) by Professor R. Barnetson; (c) reproduced from the British Society for Immunology teaching slides with permission of the Society and the Dermatology Department, London Hospital.)

(which lacks the mycobacteria), the delayed hypersensitivity state is of shorter duration and the dermal response more transient. This is known as 'Jones–Mote' sensitivity but has recently been termed **cutaneous basophil hypersensitivity** on account of the high proportion of basophils infiltrating the skin lesion.

The cellular basis of type IV hypersensitivity

Unlike the other forms of hypersensitivity which we have discussed, delayed-type reactivity cannot be transferred from a sensitized to a nonsensitized individual with serum antibody; lymphoid cells, in particular the T-lymphocytes, are required. Transfer has been achieved in the human using viable white blood cells and, interestingly, by a low molecular weight material extracted from them (Lawrence's transfer factor). The nature of this substance is, however, a mystery. The extracts contain a variety of factors which appear capable of stimulating precommitted T-cells mediating delayed hypersensitivity, but whether there is also an informational molecule

conferring antigen-specific reactivity is still a highly contentious issue.

It cannot be stressed too often that the hypersensitivity lesion results from an exaggerated interaction between antigen and the *normal* cell-mediated immune mechanisms (cf. p. 185). Following earlier priming, memory T-cells recognize the antigen together with class II major histocompatibility complex (MHC) molecules on an antigen-presenting cell and are stimulated into blast cell transformation and proliferation. The stimulated T-cells release a number of cytokines which function as mediators of the ensuing hypersensitivity response, particularly by attracting and activating macrophages if they belong to the Th1 subset, or eosinophils if they are Th2; they also help Tc precursors to become killer cells which can cause damage to virally infected target cells (figure 16.26), the CD8 TCR $\alpha\beta$ cytotoxic cells being activated by recognition of MHC class I complexes with processed viral proteins and TCR $\gamma\delta$ killers operating through binding to native viral proteins on the surface of the infected cells.

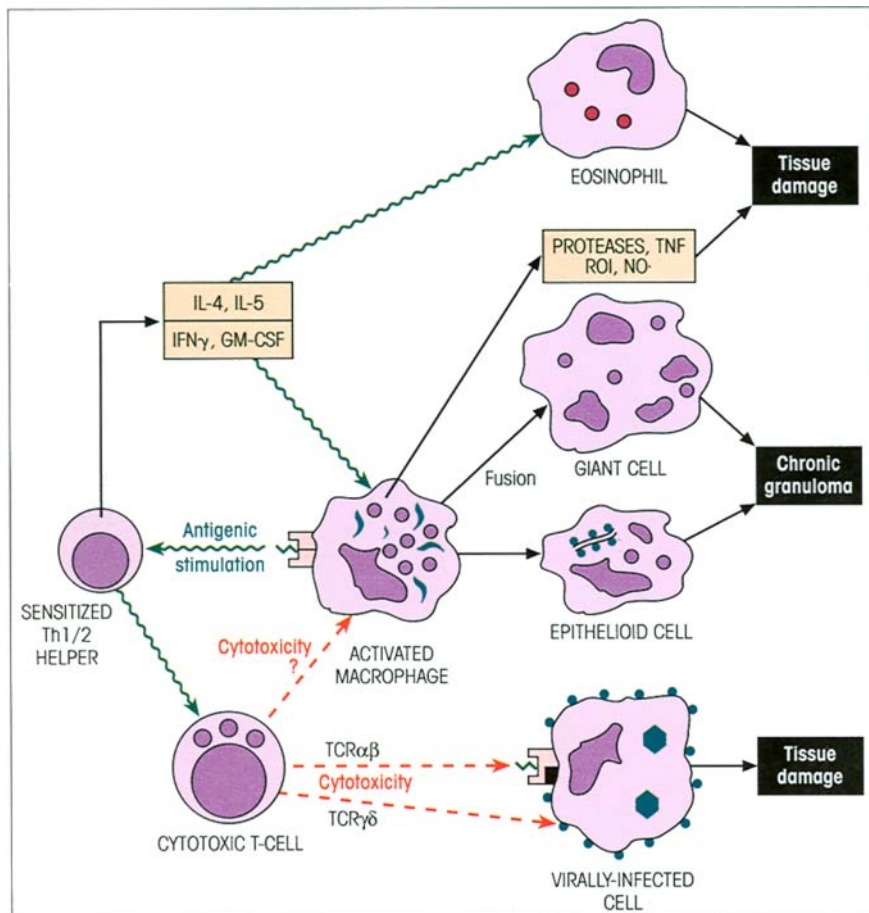


Figure 16.26. The cellular basis of type IV hypersensitivity.

Tissue damage produced by type IV reactions

Infections

The development of a state of cell-mediated hypersensitivity to bacterial products is probably responsible for the lesions associated with bacterial allergy, such as the cavitation, caseation and general toxemia seen in human tuberculosis and the granulomatous skin lesions found in patients with the borderline form of leprosy. When the battle between the replicating bacteria and the body defenses fails to be resolved in favor of the host, persisting antigen provokes a chronic local delayed hypersensitivity reaction. Continual release of cytokines from sensitized T-lymphocytes leads to the accumulation of large numbers of macrophages, many of which give rise to arrays of epithelioid cells, while others fuse to form giant cells. Macrophages bearing bacterial antigen on their surface may become targets for killer T-cells and be destroyed. Further tissue damage will occur as a result of indiscriminate cytotoxicity by lymphokine-activated macrophages (and NK cells?). Morphologically, this combination of cell types with proliferating lymphocytes and fibroblasts associated with areas of fibrosis and necrosis is termed a **chronic granuloma** and represents an attempt by the body to wall off a site of persistent infection (figures 16.25b and 16.26). It should be noted that granulomas can also arise from the persistence of indigestible antigen–antibody complexes or inorganic materials, such as talc, within macrophages, although nonimmunological granulomas may be distinguished by the absence of lymphocytes.

Crohn's disease is an **inflammatory bowel disease (IBD)** most commonly affecting the distal small intestine, and characterized by transmural inflammation with frequent granulomas but few neutrophils. The study of experimental models suggests that IBD is a dysregulated CD4 Th1-mediated immune response to one or more antigens of the indigenous colonic bacterial flora. Attempts to pin the blame on *Mycobacterium paratuberculosis* have not met with universal acceptance, although the organism can provoke a similar disease in cattle. IBD can be induced in severe combined immunodeficient mice by the transfer of CD45RB^{hi} (naive) CD4 T-cells, but this can be counter-regulated by CD4, CD25, CD45RB^{lo} cells. The aggressor cells belong to the IL-12-driven Th1 population producing TNF and IFN γ , which are highly toxic for enterocytes, whereas the regulators (still not fully characterized as Th3 and/or T-regulatory 1 [TR1] cells) secrete the suppressor cytokines TGF β and IL-10. Monoclonal anti-TNF is a very effective therapy; probiotic treatment with lactobacilli and *Streptococcus*

salivarius maintains remission in severe colitis, is less Draconian and is easier on the budget (remember the friendly yoghurt adverts). IBD induced by TCR α knockout or by administration of oxazolone presents as a relatively superficial inflammation resembling **ulcerative colitis** that is mediated by IL-4-producing Th2 cells; hence the as yet unsubstantiated observation that ulcerative colitis is a 'Th2 disease'.

The skin rashes in measles and the lesions of herpes simplex may be largely attributed to delayed-type allergic reactions with extensive damage to virally infected cells by Tc cells. By the same token, specific cytotoxic T-cells can have a field-day causing extensive destruction of liver cells infected with hepatitis B virus. Cell-mediated hypersensitivity has also been demonstrated in the fungal diseases candidiasis, dermatomycosis, coccidioidomycosis and histoplasmosis, and in the parasitic disease leishmaniasis.

Sarcoidosis is a disease of unknown etiology affecting lymphoid tissue and involving the formation of chronic granulomas. Evidence for atypical mycobacteria has been obtained, but delayed-type hypersensitivity is depressed and the patients are anergic on skin testing with tuberculin; curiously they give positive responses if cortisone is injected together with the antigen, and it has been suggested that cortisone-sensitive T-suppressors might be responsible for the anergy. The patients develop a granulomatous reaction a few weeks after intradermal injection of spleen extract from another sarcoid patient—the **Kweim reaction**.

Contact dermatitis

The epidermal route of inoculation tends to favor the development of a Th1 response through processing by class II-rich dendritic Langerhans' cells (cf. figure 2.6f) which migrate to the lymph nodes and present antigen to T-lymphocytes. Thus, delayed-type reactions in the skin are often produced by foreign low molecular weight materials capable of binding to peptides within the groove of MHC molecules on the surface of Langerhans' cells, to form new antigens. The reactions are characterized by a mononuclear cell infiltrate peaking at 12–15 hours, accompanied by edema of the epidermis with microvesicle formation (figure 16.27a and b). There is a most unusual twist to this story, however, possibly because the inciting reagent is a reactive hapten. The late mononuclear reaction is entirely dependent upon very early events (1–2 hours) mediated by hapten-specific IgM produced by B-1 cells which, together with complement, activates local vessels to permit T-cell recruitment. Contact hypersensitivity can occur in people who become sensitized while

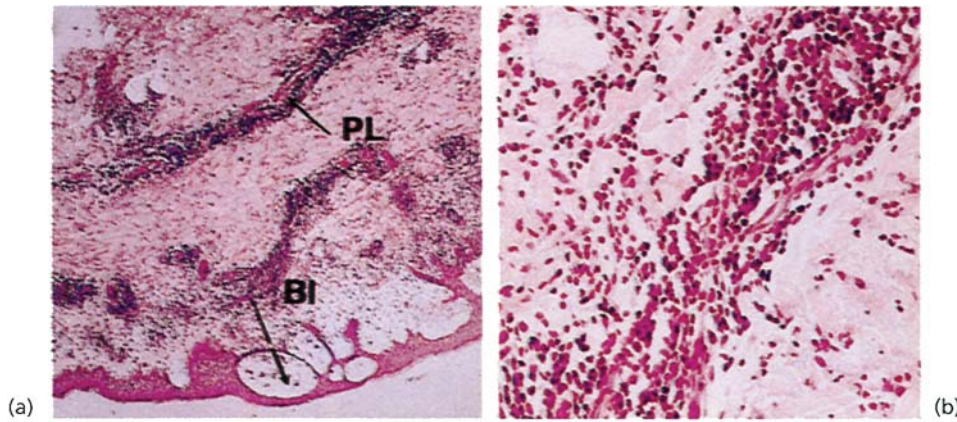


Figure 16.27. Contact sensitivity. (a) Perivascular lymphocytic infiltrates (PL) and blister (BL) formation characterize a contact sensitivity reaction of the skin. (b) High-power view to show the lymphocytic nature of the infiltrate in a contact hypersensitivity reaction. (Photographs kindly provided by Professor N. Woolf.)

working with chemicals, such as picryl chloride and chromates, or who repeatedly come into contact with the substance urushiol from the poison ivy plant. *p*-Phenylene diamine in certain hair dyes, neomycin in topically applied ointments, and nickel salts formed from articles such as nickel jewellery clasps (figure 16.25c) can provoke similar reactions. T-cell clones specific for nickel salts isolated from the latter group produce a Th1-type profile of cytokines (IFN γ , IL-2) on antigen stimulation (figure 16.9b).

Th2-mediated hypersensitivity

The examples of type IV hypersensitivity we have discussed so far are centered essentially on the Th1 macrophage pathway, but it is now clear that excessive responses by Th2 cells can damage tissues through activation of eosinophils (figure 16.26). As recounted earlier, T-cells synthesizing IL-5 are largely responsible for the sustained influx of eosinophils in asthma and atopic dermatitis (cf. p. 328). They also essentially account for the liver pathology in schistosomiasis which has been attributed to a reaction against soluble enzymes derived from the eggs which lodge in the capillaries (figure 16.28).

Other examples

Delayed hypersensitivity contributes significantly to the prolonged reactions which result from insect bites. The possible implication of allograft rejection by Tc cells as a mechanism for the control of cancer cells is discussed in Chapter 18. In certain organ-specific autoimmune diseases, such as type I diabetes, cell-mediated hypersensitivity reactions undoubtedly provide the major engine for tissue destruction.

The intestinal inflammation in celiac disease, an HLA-DQ2/8-associated enteropathy, is precipitated

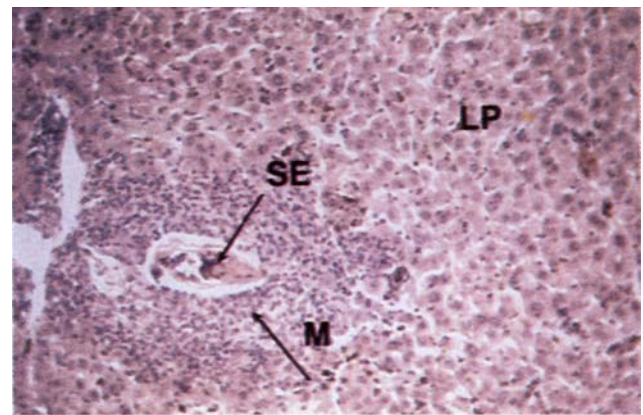


Figure 16.28. Th2-mediated response to schistosome egg. Th2-type hypersensitivity lesion of inflammatory cells (M) around a schistosome egg (SE) within the liver parenchyma (LP). (Photograph by courtesy of Professor M. Doenhoff.)

by exposure to dietary wheat gliadin. The disorder involves what is probably a genetically related increased mucosal activity of transglutaminase (the target of anti-endomysium autoantibodies; cf. p. 423). This enzyme deamidates the glutamine residues in gliadin and creates a new T-cell epitope that binds efficiently to DQ2 and is recognized by gut-derived CD4 and $\gamma\delta$ T-cells.

Psoriasis involves marked proliferation of epidermal keratinocytes and accelerated incomplete epidermal differentiation. There is inflammation in the skin with pockets of microabscesses containing neutrophils and, in all instances, CD4 cells in the psoriatic dermis and CD8 in the lesional epidermis. A reversal of epidermal dysfunction and a marked reduction in intraepidermal CD8 T-cells were achieved by systemic administration of a fusion protein of IL-2 and fragments of diphtheria toxin, which selectively blocks the growth of activated lymphocytes but not keratinocytes. This shows that the disorder is

primarily immunological rather than a dysfunction of keratinocytes which secondarily activate T-cells through, say, TGF β . The party line now would be that CD8 T-cells attack the skin, activating injury repair programs associated with wound healing, while cytokines act directly as mitogens for epidermal keratinocytes.

STIMULATORY HYPERSENSITIVITY (TYPE V)

Many cells are signaled by agents such as hormones through surface receptors to which they specifically bind the external agent, presumably through complementarity of structure. For example, when thyroid-stimulating hormone (TSH) of pituitary origin binds to the thyroid cell receptors, adenyl cyclase is activated, and the cAMP 'second messenger' which is generated acts to stimulate the thyroid cell. The **thyroid-stimulating antibody** present in the sera of thyrotoxic patients (cf. p. 422) is an autoantibody directed against a site on the TSH receptor which produces the changes required for adenyl cyclase activation. Intriguingly, but not much more at present, there are indications that cimetidine-resistant duodenal ulcer patients might have stimulatory antibodies directed to histamine receptors.

Experimental examples of stimulation by antibodies to cell surface antigens may be cited: the activation of B- and T-lymphocytes by F(ab')₂ fragments directed to their antigen-specific receptors; the production of cell division in thyroid cells by 'growth' autoantibodies; the induction of pinocytosis by antimacrophage serum; and the mitogenic effect of antibodies to sea-urchin eggs. It is worthy of note that, although antibodies to enzymes directed against determinants near to the active site can exert a blocking effect, combination with more distant determinants can sometimes bring about allosteric conformational changes which are associated with a considerable increase in enzymic activity, as has been described for certain variants of penicillinase and β -galactosidase.

'INNATE' HYPERSENSITIVITY REACTIONS

Many infections provoke a '**toxic shock syndrome**' characterized by hypotension, hypoxia, oliguria and microvascular abnormalities and mediated by elements of the innate immune system independently of the operation of acquired immune responses.

Septicemia associated with **Gram-negative bacteria** results in excessive release of TNF, IL-1 and IL-6 through stimulation of macrophages and endothelial cells by the lipopolysaccharide (LPS) endotoxin

(cf. figure 1.6). Normally this would enhance host defenses, aiding the recruitment of phagocytes by promoting adherence to endothelium, priming neutrophils for subsequent release of reactive oxygen intermediates, inducing febrile responses (immune responses improve steadily from 33 to 44°C), and so on. Unfortunately, the excess of circulating LPS and the cytokines released by it lead to unwanted pathophysiology at distant sites, such as the **adult respiratory distress syndrome** brought about by an overwhelming invasion of the lung by neutrophils. There is a prolonged pathologically high concentration of nitric oxide but, additionally, LPS can activate the alternative complement pathway, and this may be linked to its ability to induce the release of thromboxane A₂ and prostaglandin from platelets leading to **disseminated intravascular coagulation**.

Whereas the major culprit in Gram-negative sepsis is LPS, **Gram-positive organisms** possess a variety of components which act on host defense elements to initiate septic shock. Thus adherence of *Staphylococcus aureus* to macrophages induces TNF α synthesis, and peptidoglycan-mediated aggregation of platelets by the same organism leads to disseminated intravascular coagulation. The staphylococcal and streptococcal enterotoxins induce toxic shock syndrome by quite different means. By functioning as **superantigens** (cf. p. 103), they react directly with particular T-cell receptor families and give rise to massive cytokine release, including TNF and macrophage migration inhibitory factor (MIF), which is detected in high concentrations in the plasma of patients with septic shock. Various treatments are under investigation. Pentoxifylline blocks TNF production by macrophages. Experimental models of septic shock can be blocked by anti-MIF and by a peptide derived from the natural sequence 150–161 of staphylococcal enterotoxin B, which is part of a domain crucial for T-cell activation.

Germ-line mutations in the extracellular domains of the 55 kDa TNF receptor, TNFR1, define a family of dominantly inherited autoinflammatory syndromes involving unpredictable periodic fever episodes.

The reader's attention has already been drawn to the unusual susceptibility of erythrocytes to lysis in **paroxysmal nocturnal hemoglobinuria** resulting from deficiency in complement control proteins on the red cell surface (see p. 307). Undue C3 consumption is associated with mesangiocapillary glomerulonephritis and partial lipodystrophy in patients with the so-called **C3 nephritic factor**, which appears to be an IgG autoantibody capable of activating the alternative pathway by combining with and stabilizing the C3bBb convertase.

We should include **idiopathic pulmonary fibrosis** in this section. It is a chronic fatal disorder characterized by diffuse fibrosis of the alveolar walls in which local macrophages play a central role. On activation, they produce an early G1 competence growth signal, such as platelet-derived growth factor and fibronectin, and then a late G1 progression growth signal, such as insulin-like growth factor 1. As a result, the fibroblasts multiply and become embedded in a collagen matrix to the respiratory detriment of the host.

The neuropathological hallmarks of Alzheimer's

disease are extracellular plaques and intracellular neurofibrillary tangles. One of the constituents of senile plaques is β -amyloid composed of a hydrophobic fragment of amyloid precursor protein (APP). Normally APP is cleaved by an α -secretase into soluble products which cannot form β -amyloid, but members of a family diagnosed with Alzheimer's disease, and having a pathogenic mutation at residues 670–671 of APP, had very low levels of this soluble cleaved form compared with noncarriers. This could also represent a new and promising diagnostic marker.

SUMMARY

- Excessive stimulation of the normal effector mechanisms of the immune system can lead to tissue damage and we speak of hypersensitivity reactions, several types of which can be distinguished.

Anaphylactic hypersensitivity (type I)

- Anaphylaxis involves contraction of smooth muscle and dilatation of capillaries.
- This depends upon the reaction of antigen with specific IgE antibody bound through its Fc to the mast cell high affinity receptor Fc ϵ RI.
- Cross-linking and clustering of the IgE receptors activates the Lyn protein tyrosine kinase, recruits other kinases and leads to release from the granules of mediators including histamine, leukotrienes and platelet activating factor, plus eosinophil and neutrophil chemotactic factors and the cytokines IL-3, -4, -5 and GM-CSF.

Atopic allergy

- Atopy stems from an excessive IgE response to extrinsic antigens (allergens) which leads to local anaphylactic reactions at sites of contact with allergen.
- Hay fever and extrinsic asthma represent the most common atopic allergic disorders resulting from exposure to inhaled allergens. Atopic dermatitis is also extremely common.
- Whereas the immediate reaction to extrinsic allergen (maximum at 30 minutes) is due to mast cell triggering, a late phase reaction peaking at 5 hours, involving heavy eosinophil infiltration, is initiated by the activation of alveolar and other macrophages through surface-bound IgE; secreted TNF and IL-1 β now act upon epithelial cells and fibroblasts to release powerful eosinophil chemoattractants such as RANTES and eotaxin.
- In asthma, serious prolongation of the response to allergen is caused by T-cells of Th2 type which sustain the re-

cruitment of tissue-damaging eosinophils through the release of IL-5. The soup of powerful bronchoconstrictors, the injurious effect of eosinophil major basic protein and reactive oxygen intermediates and the mucus hypersecretion stimulated by IL-13 and IL-4, all contribute to the drastic airway damage characteristic of chronic asthma.

- Many food allergies involve type I hypersensitivity.
- Strong genetic factors include linkage to the propensity to make the IgE isotype and to genes involving the IL-4/9/13, GM-CSF and CD14 cluster.
- Exposure to certain foods and to Th1-stimulating infections may strongly influence the 'immunostat' setting of the tendency to either Th1 or Th2 responses, the latter increasing the risk of allergy through promotion of IgE synthesis and eosinophil recruitment.
- The offending antigen is identified by intradermal prick tests, giving immediate wheal and erythema reactions, by provocation testing and by RAST.
- Where possible, allergen avoidance is the best treatment.
- A monoclonal antibody directed to the receptor-binding domain of IgE dramatically reduces IgE levels and synthesis, and decreases mast cell responsiveness. Symptomatic treatment involves the use of long-acting β_2 -agonists and newly developed leukotriene antagonists. Chromones, such as sodium cromoglycate, block chloride channel activity thereby stabilizing mast cells and inhibiting bronchoconstriction. Theophylline, the single most prescribed drug for asthma, is a phosphodiesterase inhibitor which raises intracellular calcium; this causes bronchodilatation and inhibition of IL-5 effects on eosinophils. Chronic asthma is dominated by activated Th2 cells and is treated with topical steroids which display a wide range of anti-inflammatory actions, including the ability to block the production of mediators by

(continued)

stimulated macrophages or Th2 cells. These are supplemented where necessary by long-acting β_2 -agonists and theophylline.

- Courses of antigen injection may desensitize by the formation of blocking, or possibly regulatory, IgG antibodies or through T-cell regulation. T-cell epitope peptides may be manipulated to modulate the atopic state.

Antibody-dependent cytotoxic hypersensitivity (type II)

- This involves the death of cells bearing antibody attached to a surface antigen.
- The cells may be taken up by phagocytic cells to which they adhere through their coating of IgG or C3b or lysed by the operation of the full complement system.
- Cells bearing IgG may also be killed by polymorphs and macrophages or by K-cells through an extracellular mechanism (antibody-dependent cellular cytotoxicity).
- Examples are: transfusion reactions, hemolytic disease of the newborn through rhesus incompatibility, antibody-mediated graft destruction, autoimmune reactions directed against the formed elements of the blood and kidney glomerular basement membranes, and hypersensitivity resulting from the coating of erythrocytes or platelets by a drug.

Complex-mediated hypersensitivity (type III)

- This results from the effects of antigen–antibody complexes through (i) activation of complement and attraction

of polymorphonuclear leukocytes which release tissue-damaging mediators on contact with the complex, (ii) stimulation of macrophages to release proinflammatory cytokines, and (iii) aggregation of platelets to cause microthrombi and vasoactive amine release.

- Where circulating antibody levels are high, the antigen is precipitated near the site of entry into the body. The reaction in the skin is characterized by polymorph infiltration, edema and erythema maximal at 3–8 hours (Arthus reaction).

- Examples are farmer's lung, pigeon-fancier's disease and pulmonary aspergillosis where inhaled antigens provoke high antibody levels, reactions to an abrupt increase in antigen caused by microbial cell death during chemotherapy for leprosy or syphilis, polyarteritis nodosa linked to complexes with hepatitis B virus and an element of the synovial lesion in rheumatoid arthritis.

- In relative *antigen excess*, soluble complexes are formed which are removed by binding to the CR1 C3b receptors on red cells. If this system is overloaded or if the classical complement components are deficient, the complexes circulate in the free state and are deposited under circumstances of increased vascular permeability at certain preferred sites: the kidney glomerulus, the joints, the skin and the choroid plexus.

- Examples are: glomerulonephritis associated with systemic lupus erythematosus (SLE) or infections with streptococci, malaria and other parasites, neurological dis-

Table 16.3. Comparison of types of hypersensitivity involving acquired responses.

	Anaphylactic (I)	Cytotoxic (II)	Complex-mediated (III)	Cell-mediated (IV)	Stimulatory (V)
Antibody mediating reaction	Homocytotropic Ab Mast-cell binding	Humoral Ab ± CF*	Humoral Ab ± CF*	Receptor on T-lymphocyte	Humoral Ab Non-CF*
Antigen	Usually exogenous (e.g. grass pollen)	Cell surface	Extracellular	Associated with MHC on macrophage or target cell	Cell surface
Response to intradermal antigen:	Max. reaction Appearance Histology	30 min (+ late reaction) Wheal and flare Degranulated mast cells; edema; (late reaction cellular including eosinophils)	3–8 h Erythema and edema Acute inflammatory reaction; pre-dominant polymorphs	24–48 h Erythema and induration Perivascular inflammation: polymorphs migrate out leaving predominantly mono-nuclear cells	— — —
Transfer sensitivity to normal subject	←	Serum antibody	→	Lymphoid cells Transfer factor	Serum antibody
Examples:	Atopic allergy, e.g. hay fever	Hemolytic disease of newborn (Rh)	Complex glomerulonephritis Farmer's lung	Mantoux reaction to TB Granulomatous reaction to TB Contact sensitivity	Thyrotoxicosis

*CF, complement fixation.

turbances in SLE and subacute sclerosing panencephalitis, and hemorrhagic shock in dengue viral infection.

Cell-mediated or delayed-type hypersensitivity (type IV)

- This is based upon the interaction of antigen with primed T-cells and represents tissue damage resulting from inappropriate cell-mediated immunity reactions.
- A number of soluble cytokines including IFN γ are released, which activate macrophages and account for the events that occur in a typical delayed hypersensitivity response such as the Mantoux reaction to tuberculin, that is, the delayed appearance of an indurated and erythematous reaction which reaches a maximum at 24–48 hours and is characterized histologically by infiltration with mononuclear phagocytes and lymphocytes.
- Continuing provocation of delayed hypersensitivity by persisting antigen leads to the formation of chronic granulomas.
- Th2-type cells producing IL-4 and IL-5 can also produce tissue damage through their ability to recruit eosinophils.
- CD8 T-cells are activated by class I major histocompatibility antigens to become directly cytotoxic to target cells bearing the appropriate antigen.
- Examples are: tissue damage occurring in bacterial (tuberculosis, leprosy), viral (measles, herpes), fungal (candidiasis, histoplasmosis) and parasitic (leishmaniasis, schistosomiasis) infections, contact dermatitis from exposure to chromates and poison ivy, insect bites and psoriasis. Inflammatory bowel disease can result from Th1-type

reactions to intestinal bacteria or certain foods such as wheat gliadin.

Stimulatory hypersensitivity (type V)

- The antibody reacts with a key surface component such as a hormone receptor and 'switches on' the cell.
 - An example is the thyroid hyper-reactivity in Graves' disease due to a thyroid-stimulating autoantibody.
- Features of these five types of acquired hypersensitivity are compared in table 16.3.

'Innate' hypersensitivity reactions

- Many infections provoke a 'toxic shock syndrome' involving excessive release of TNF, IL-1 and IL-6 and intravascular activation of complement.
- Septic shock associated with Gram-negative bacteria is primarily due to the lipopolysaccharide (LPS) endotoxin.
- Gram-positive organisms cause release of TNF and macrophage migration inhibitory factor (MIF) through direct action on macrophages and stimulation of selected T-cell families by the enterotoxin superantigens. Aggregation of platelets by *S. aureus* induces disseminated intravascular coagulation.
- Aberration of innate mechanisms may underlie idiopathic pulmonary fibrosis and contribute to the β -amyloid plaques in Alzheimer's disease.

See the accompanying website (www.roitt.com) for multiple choice questions.

FURTHER READING

- Arad G. *et al.* (2000) Superantigen antagonist protects against lethal shock and defines a new domain for T-cell activation. *Nature Medicine* **6**, 414.
- Chapel H., Haeney M., Misbah S. & Snowden N. (1999) *Essentials of Clinical Immunology*, 4th edn. Blackwell Science, Oxford. (Very broad account of the diseases involving the immune system. Good illustration by case histories and the laboratory tests available. Also MCQs.)
- Coleman J.W. & Blanca M. (1998) Mechanisms of drug allergy. *Immunology Today* **19**, 196.
- Holgate S. (ed.) (1999) Allergy and asthma. *Nature* **402** (Suppl.) (Excellent reviews on genetic factors, the role of IgE in the development of asthma, therapeutic strategies, and an especially valuable account of IgE receptor Fc ϵ RI signaling.)
- Holgate S.T. & Church M.K. (1993) *Allergy*. Gower Medical Publishing, London.
- Kinet J.-P. (ed.) (1999) Atopy and other hypersensitivities. *Current Opinion in Immunology* **11**, 603. (A collection of critical reviews on hypersensitivity which appear annually as Issue No. 6 and are highly recommended.)
- Molberg O. *et al.* (1998) Tissue transglutaminase selectively modifies gliadin peptides that are recognized by gut-derived T-cells in celiac disease. *Nature Medicine* **4**, 713.
- Ravetch J.V. (1998) Uncoupling immune complex formation and kidney damage in autoimmune glomerulonephritis. *Science* **279**, 1052.
- Scadding G. (1998) Atopic allergy. In Delves P.J. & Roitt I.M. (eds) *Encyclopedia of Immunology*, 2nd edn, p. 251. Academic Press, London.
- Tsuji R.F. *et al.* (1997) Required early complement activation in contact sensitivity with generation of local C5-dependent chemotactic activity, and late T-cell interferon gamma: a possible initiating role of B-cells. *Journal of Experimental Medicine* **186**, 1015.
- Wills-Karp M. (1999) Immunologic basis of antigen-induced airway hyperresponsiveness. *Annual Review of Immunology* **17**, 255.

Transplantation

Introduction, 349

Genetic control of transplantation antigens, 349

Some other consequences of MHC incompatibility, 352

Class II MHC differences produce a mixed lymphocyte reaction (MLR), 352

The graft-vs-host (g.v.h.) reaction, 352

Mechanisms of graft rejection, 352

Lymphocytes can mediate rejection, 352

The allograft response is powerful, 353

The role of humoral antibody, 354

The prevention of graft rejection, 355

Matching tissue types on graft donor and recipient, 355

Agents producing general immunosuppression, 357

Strategies for antigen-specific depression of allograft reactivity, 360

Is xenografting a practical proposition?, 361

Tissue engineering, 362

Clinical experience in grafting, 362

Privileged sites, 362

Kidney grafts, 362

Heart transplants, 363

Liver transplants, 364

Bone marrow grafting, 364

Other organs, 365

Association of HLA type with disease, 366

Linkage disequilibrium and disease susceptibility, 366

Association with immunological diseases, 367

Reproductive immunology, 369

The fetus is a potential allograft, 369

Fertility can be controlled immunologically, 370

Summary, 371

Further reading, 372

INTRODUCTION

The replacement of diseased organs by a transplant of healthy tissue has long been an objective in medicine but has been frustrated to no mean degree by the uncooperative attempts by the body to reject grafts from other individuals. Before discussing the nature and implications of this rejection phenomenon, it would be helpful to define the terms used for transplants between individuals and species.

Autograft—tissue grafted back on to the original donor.

Isograft—graft between **syngeneic** individuals (i.e. of identical genetic constitution) such as identical twins or mice of the same pure line strain.

Allograft (old term, homograft)—graft between **allogeneic** individuals (i.e. members of the same species but different genetic constitution), e.g. human to human and one mouse strain to another.

Xenograft (heterograft)—graft between **xenogeneic** individuals (i.e. of different species), e.g. pig to human.

It is with the allograft reaction that we have been most concerned, although there is now a serious interest in the use of grafts from other species. The most common allografting procedure is probably

blood transfusion where the unfortunate consequences of mismatching are well known. Considerable attention has been paid to the rejection of solid grafts such as skin and the sequence of events is worth describing. In mice, for example, the skin allograft settles down and becomes vascularized within a few days. Between 3 and 9 days the circulation gradually diminishes and there is increasing infiltration of the graft bed with lymphocytes and monocytes but very few plasma cells. Necrosis begins to be visible macroscopically and within a day or so the graft is sloughed completely (figure M17.1.1). **Rejection has all the hallmarks of an immunological response** in that it shows both memory and specificity (Milestone 17.1). Furthermore, the recipient of T-cells from a donor which has already rejected a graft will give accelerated rejection of a further graft of the same type (figure 17.1), showing that the lymphoid cells are primed and retain memory of the first contact with graft antigens.

GENETIC CONTROL OF TRANSPLANTATION ANTIGENS

The specificity of the antigens involved in graft rejection is under genetic control. Genetically identical in-

Milestone 17.1—The Immunological Basis of Graft Rejection

The field of transplantation owes a tremendous debt to Sir Peter Medawar, the outstanding scientist who kick-started and inspired its development. Even at the turn of the century it was an accepted paradigm that grafts between unrelated members of a species would be unceremoniously rejected after a brief initial period of acceptance (figure

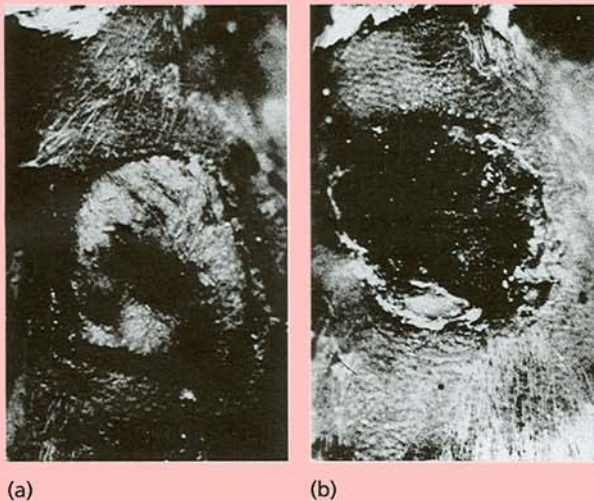


Figure M17.1.1. Rejection of CBA skin graft by strain A mouse. (a) Ten days after transplantation; discolored areas caused by destruction of epithelium and drying of the exposed dermis. (b) Thirteen days after transplantation; the scabby surface indicates total destruction of the graft. (Photographs courtesy of Professor L. Brent.)

M17.1.1). That there was an underlying genetic basis for rejection became apparent from Padgett's observations in Kansas City in 1932 that skin allografts between family members tended to survive for longer than those between unrelated individuals and J.B. Brown's critical demonstration in St Louis in 1937 that monozygotic (i.e. genetically identical) twins accepted skin grafts from each other. However, it was not until Medawar's research in the early part of the Second World War, motivated by the need to treat aircrew with appalling burns, that rejection was laid at immunology's door. He showed that a second graft from a given donor was rejected more rapidly and more vigorously than the first and, further, that an unrelated graft was rejected with the kinetics of a first set reaction (figure M17.1.2). This **second set rejection** is characterized by **memory** and **specificity** and thereby bears the hallmarks of an immunological response. This of course was later confirmed by transferring the ability to express a second set reaction with lymphocytes.

The message was clear: to achieve successful transplantation of tissues and organs in the human, it would be necessary to overcome this immunogenetic barrier. Limited success was obtained by Murray at the Peter Bent Brigham Hospital and Hamburger in Paris, who grafted kidneys between dizygotic twins using sublethal X-irradiation. The key breakthrough came when Schwartz and Damashek's report on the immunosuppressive effects of the antimitotic drug 6-mercaptopurine was applied inde-

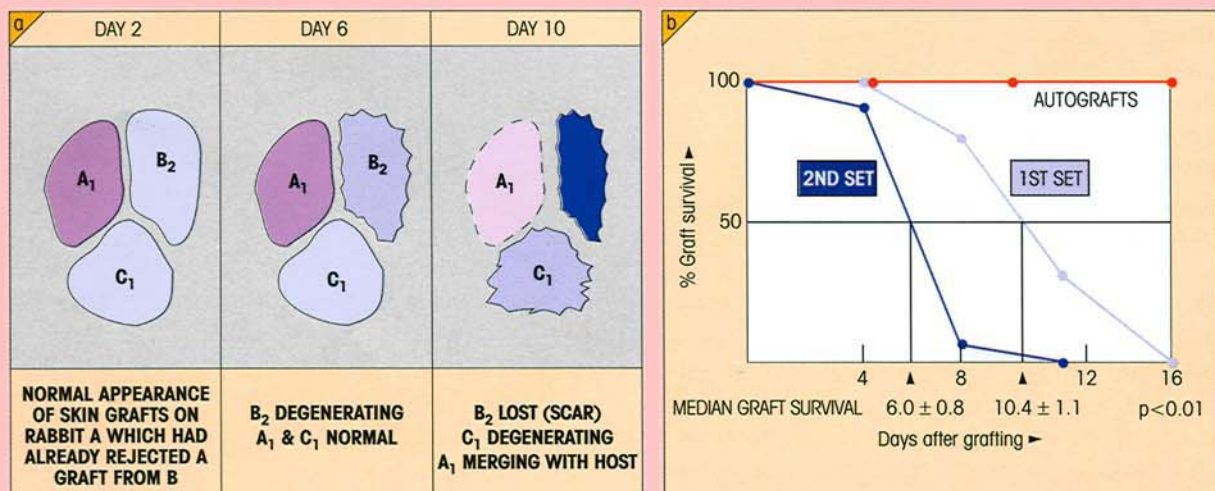


Figure M17.1.2. Memory and specificity in skin allograft rejection in rabbits. (a) Autografts and allografts from two unrelated donors B and C are applied to the thoracic wall skin of rabbit A which has already rejected a first graft from B (B₁). While the autograft A remains intact, graft C₁ seen for the first time undergoes 1st

set rejection, whereas a *second* graft from B (B₂) is sloughed off very rapidly. (b) Median survival times of 1st and 2nd set skin allografts showing faster 2nd set rejection. (From Medawar P.B. (1944) The behavior and fate of skin autografts and skin homografts (allografts) in rabbits. *Journal of Anatomy* 78, 176.)

(continued)

pendently by Calne and Zukowski in 1960 to the prolongation of renal allografts in dogs. This was followed very rapidly by Murray's successful grafting in 1962 of an unrelated cadaveric kidney under the immunosuppressive umbrella of azathioprine, the more effective derivative of 6-mercaptopurine devised by Hutchings and Elion.

This story is studded with Nobel Prize winners and

readers of a historical bent will gain further insight into the development of this field and the minds of the scientists who gave medicine this wonderful prize in Terasaki P.I. (ed.) (1991) *History of Transplantation; Thirty-Five Recollections*, UCLA Tissue Typing Laboratory, Los Angeles, CA and, subsequently, Brent L. (1996) *A History of Transplantation Immunology*, Academic Press, London.

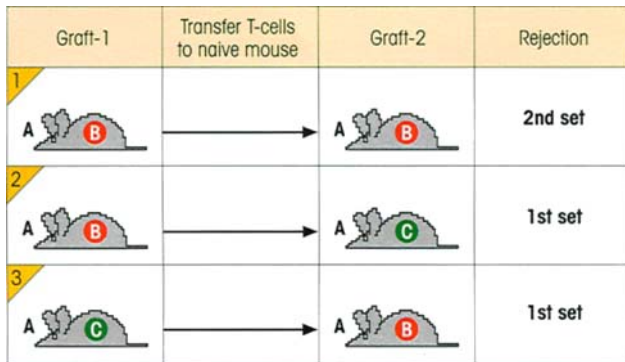


Figure 17.1. Graft rejection induces memory which is specific and can be transferred by T-cells. In experiment 1, an A strain recipient of T-cells from another A strain mouse, which had rejected a graft from strain B, will give accelerated (i.e. 2nd set) rejection of a B graft. Experiments 2 and 3 show the specificity of the phenomenon with respect to the genetically unrelated third party strain C.

dividuals, such as mice of a pure strain or monozygotic twins, have identical transplantation antigens and grafts can be freely exchanged between them. The Mendelian segregation of the genes controlling these antigens has been revealed by interbreeding experiments between mice of different pure strains. Since these mice breed true within a given strain and always accept grafts from each other, they must be homozygous for the 'transplantation' genes. Consider two such strains A and B with allelic genes differing at one locus. In each case paternal and maternal genes will be identical and they will have a genetic constitution of, say, A/A and B/B respectively. Crossing strains A and B gives a first familial generation (F1) of constitution A/B . Now, all F1 mice accept grafts from either parent showing that they are immunologically tolerant to both A and B due to the fact that the transplantation antigens from each parent are codominantly expressed (figure 4.17). By intercrossing the F1 generation, one would expect an average distribution of genotypes for the F2s as shown in figure 17.2; only one in four would have no A genes and would therefore reject an A graft because of lack of tolerance, and one in four would re-

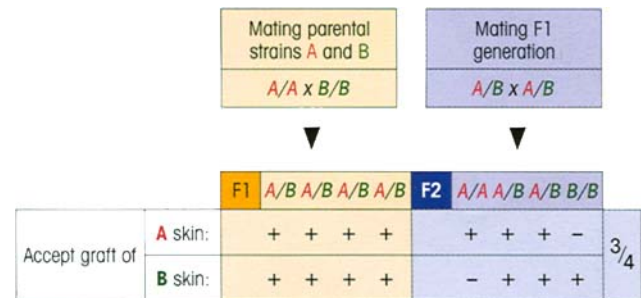


Figure 17.2. Inheritance of genes controlling transplantation antigens. A represents a gene expressing the A antigen and B the corresponding allelic gene at the same genetic locus. The pure strains are homozygous for A/A and B/B respectively. Since the genes are codominant, an animal with A/B genome will express both antigens, become tolerant to them and therefore accept grafts from either A or B donors. The illustration shows that, for each gene controlling a transplantation antigen specificity, three-quarters of the F2 generation will accept a graft of parental skin. For n genes the fraction is $(3/4)^n$. If F1 A/B animals are back-crossed with an A/A parent, half the progeny will be A/A and half A/B ; only the latter will accept B grafts.

ject B grafts for the same reason. Thus, for each locus, three out of four of the F2 generation will accept parental strain grafts.

In the mouse around 40 such loci have been established but, as we have seen earlier, the complex locus termed H2 (HLA in the human) predominates in the sense that it controls the 'strong' transplantation antigens which provoke intense allograft reactions. We have looked at the structure (cf. figure 4.11) and biology of this **major histocompatibility locus** in some detail in previous chapters (see Milestone 4.2, p. 71). The non-H-2 or 'minor' transplantation antigens, such as the male H-Y, are recognized as processed peptides in association with the *major histocompatibility complex* (MHC) molecules on the cell surface by T-cells but not at all by B-cells. One should not be misled by the term 'minor' into thinking that these antigens do not give rise to serious rejection problems, albeit more slowly than the MHC.

SOME OTHER CONSEQUENCES OF MHC INCOMPATIBILITY

Class II MHC differences produce a mixed lymphocyte reaction (MLR)

When lymphocytes from individuals of different class II haplotype are cultured together, blast cell transformation and mitosis occur (MLR), the T-cells of each population of lymphocytes reacting against MHC class II determinants on the surface of the other population. The responding cells belong predominantly to a population of CD4⁺ T-lymphocytes and are stimulated by the class II determinants present mostly on B-cells, macrophages and especially dendritic antigen-presenting cells. Thus, the MLR is inhibited by antisera to class II determinants on the stimulator cells.

The graft-vs-host (g.v.h.) reaction

When competent T-cells are transferred from a donor to a recipient which is incapable of rejecting them, the grafted cells survive and have time to recognize the host antigens and react immunologically against them. Instead of the normal transplantation reaction of host against graft, we have the reverse, the so-called graft-vs-host (g.v.h.) reaction. In the young rodent there can be inhibition of growth (runtling), spleen enlargement and hemolytic anemia (due to the production of red cell antibodies). In the human, fever, anemia, weight loss, rash, diarrhea and splenomegaly are observed, with cytokines, especially tumor necrosis factor (TNF), being thought to be the major mediators of pathology. The 'stronger' the transplantation antigen difference, the more severe the reaction. Where donor and recipient differ at HLA or H-2 loci, the consequences can be fatal, although it should be noted that reactions to dominant minor transplantation antigens, or combinations of them, may be equally difficult to control.

Two possible situations leading to g.v.h. reactions are illustrated in figure 17.3. In the human this may arise in immunologically anergic subjects receiving bone marrow grafts, e.g. for combined immunodeficiency (see p. 312), for red cell aplasia after radiation accidents, or as a possible form of cancer therapy. Competent T-cells in blood or present in grafted organs given to immunosuppressed patients may give g.v.h. reactions; so could maternal cells which adventitiously cross the placenta, although in this case there is as yet no evidence of diseases caused by such a mechanism in the human.

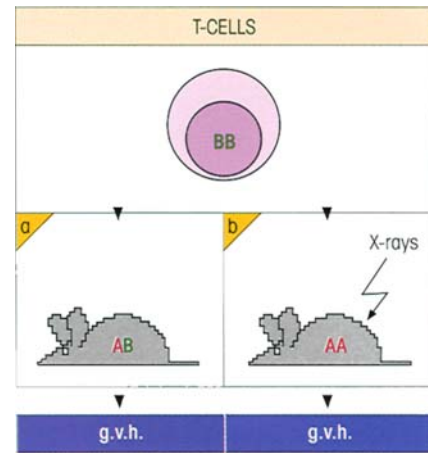


Figure 17.3. Graft-vs-host reaction. When competent T-cells are inoculated into a host incapable of reacting against them, the grafted cells are free to react against the antigens on the host's cells which they recognize as foreign. The ensuing reaction may be fatal. Two of many possible situations are illustrated: (a) the hybrid AB receives cells from one parent (BB) which are tolerated but react against the A antigen on host cells; (b) an X-irradiated AA recipient restored immunologically with BB cells cannot react against the graft and a g.v.h. reaction will result.

MECHANISMS OF GRAFT REJECTION

Lymphocytes can mediate rejection

A great deal of the work on allograft rejection has involved transplants of skin or solid tumors because their fate is relatively easy to follow. In these cases there is little support for the view that humoral antibodies are instrumental in destruction of the graft, although, as we shall see later, this is not necessarily so with transplants of other organs such as the kidney. Whereas passive transfer of *serum* from an animal which has rejected a skin allograft cannot usually accelerate the rejection of a similar graft on the recipient animal, injection of *lymphoid* cells (particularly recirculating small lymphocytes) is effective in shortening graft survival (cf. figure 17.1).

A primary role of lymphoid cells in first set rejection would be consistent with the histology of the early reaction showing infiltration by mononuclear cells with very few polymorphs or plasma cells (figure 17.4). The dramatic effect of neonatal thymectomy on prolonging skin transplants, as mentioned earlier, and the long survival of grafts on children with thymic deficiencies implicate the T-lymphocytes in these reactions. In the chicken, homograft rejection and g.v.h. reactivity are influenced by neonatal thymectomy but not bursectomy. More direct evidence has come from *in vitro* studies showing that T-cells taken from mice rejecting

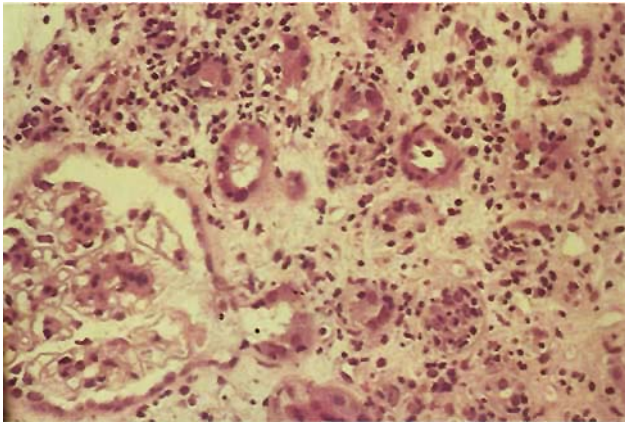


Figure 17.4. Acute rejection of human renal allograft showing dense cellular infiltration of interstitium by mononuclear cells. (Photograph courtesy of Drs M. Thompson and A. Dorling.)

an allograft could kill target cells bearing the graft antigens *in vitro*. Recent work on the importance of murine and human CD4⁺ cells as effectors has cast some doubt, probably wrongly, on the role of CD8 cytotoxic cells in graft rejection *in vivo*; although sometimes CD4 cells have cytotoxic potential for class II targets, as a rule they are associated with helper activity, in this case particularly for Tc precursors, and with the production of cytokines mediating delayed hypersensitivity reactions. Perhaps they act to encourage access of Tc cells to their targets? We do know that γ -interferon (IFN γ) up-regulates antigen expression on the target graft cell, so increasing its vulnerability to CD8 cytotoxic cells.

The allograft response is powerful

Remember, we defined the MHC by its ability to provoke the most powerful rejection of grafts between members of the same species. This intensity of MHC mismatched rejection is a consequence of the **very high frequency of alloreactive cells** (i.e. cells which react with allografts) **present in normal individuals**. Whereas merely a fraction of a per cent of the normal T-cell population is specific for a given single peptide, upwards of 10% of the T-cells react with alloantigens. This remarkable phenomenon seems to be based largely on the ability of **alloreactive T-cells to recognize allo- (i.e. graft) MHC plus self-peptides** (figure 17.5a). The allogeneic MHC differs from the recipient essentially in the groove residues which contact processed peptide, but much less so in the more conserved helical regions which are recognized by the TCR. Having a different groove structure, the allo-MHC will be able to bind a number of peptides derived from proteins common to donor and host which might be unable to fit the

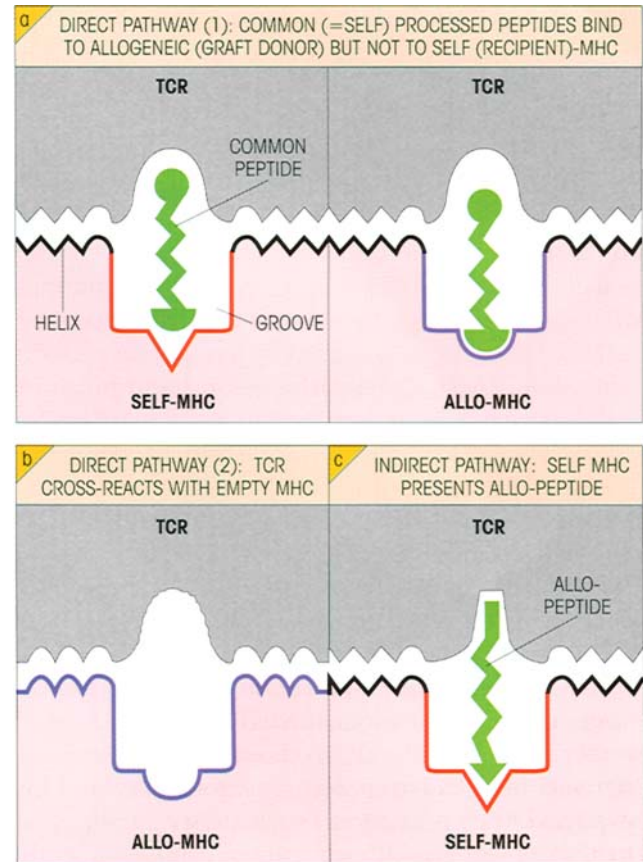


Figure 17.5. Recognition of graft antigens by alloreactive T-cells. **Direct pathway.** (a) Polymorphic differences largely affect peptide binding but not T-cell receptor (TCR) contact by the donor MHC. Under these circumstances, the donor MHC molecule will be seen as 'self' by the host T-cells but, unlike the self-MHC, the donor MHC groove on graft antigen-presenting cells will bind large numbers of processed serum and cellular peptides common to graft and recipient to which the responder host T-cells have not been rendered tolerant and which can therefore provoke a reaction in up to 10% of these host T-cells. This provides the intensity of the allograft response. This explanation for the high frequency of alloreactive T-cells is given further credibility by the isolation of individual T-cell clones which react with self- and allo-MHC, each binding a different peptide sequence. (b) Minor polymorphic residue changes in the α -helix may adventitiously allow binding of TCRs to the allo-MHC independently of the associated peptide. Multiple bonds of this nature between the APC and T-cell may give rise to a strong enough interaction to permit T-cell activation. **Indirect pathway.** (c) Polymorphic peptides derived from the graft may be presented by self-MHC on host antigen-presenting cells to an initially small population of T-cells which will expand with time. Regions which differ from self-MHC with respect to polymorphic residues are indicated in blue.

groove in the host MHC and therefore fail to induce self-tolerance. Thus the host T-cells which recognize allo-MHC plus common peptides will not have been eliminated, and will be available to react with the large number of different peptides binding to the allo-groove of the donor antigen-presenting cells (APCs)

which migrate to the secondary lymphoid tissue of the graft recipient. In some cases, the polymorphic residues may lie within the regions of the MHC helices which contact TCR directly and, by chance, a **proportion of the T-cell repertoire cross-reacts and binds to the donor MHC with high affinity** (figure 17.5b). Attachment of the T-cell to the APC will be particularly strong since the TCRs will bind to all the donor MHC molecules on the APC, whereas, in the case of normal MHC–peptide recognition, only a small proportion of the MHC grooves will be filled by the specific peptide in question. These **direct pathways of immunization** by the allograft MHC which are usually initiated by the most powerful APC, the dendritic cell, dominate the early sensitization events, since this acute phase of rejection (see below) can be blocked by antibodies to the *allo*-MHC class II.

However, with time, as the donor APCs in the graft are replaced by recipient cells, another rejection mechanism based on an **indirect pathway of sensitization** involving the **presentation of processed allogeneic peptides by host MHC** (figure 17.5c) becomes possible. Although T-cells recognizing peptides derived from polymorphic graft proteins would be expected to be present in low frequency comparable to that observed with any foreign antigen, a graft which has been in place for an extended period will have the time to expand this small population significantly so that later rejection may depend progressively on this indirect pathway. In these circumstances, *anti-recipient* MHC class II can now be shown to prolong renal allografts in rats.

The role of humoral antibody

It has long been recognized that isolated allogeneic cells such as lymphocytes can be destroyed by cytotoxic (type II) reactions involving humoral antibody. However, although earlier experience with skin and solid tumor grafts suggested that they were not readily susceptible to the action of cytotoxic antibodies, it is now clear that this does not hold for all types of organ transplants. Consideration of the different ways in which kidney allografts can be rejected illustrates the point.

Hyperacute rejection within minutes of transplantation, characterized by sludging of red cells and microthrombi in the glomeruli, occurs in individuals with pre-existing humoral antibodies—either due to blood group incompatibility or presensitization to class I MHC through blood transfusion.

Acute early rejection occurring up to 10 days or so after transplantation is characterized by dense cellular

infiltration (figure 17.4) and rupture of peritubular capillaries, and appears to be a cell-mediated hypersensitivity reaction mainly involving CD8 cytotoxic attack on graft cells whose MHC antigen expression has been upregulated by γ -interferon.

Acute late rejection, which occurs from 11 days onwards in patients suppressed with prednisolone and azathioprine, may be due to breakthrough of immunosuppression by the immune response, or can be caused by the binding of immunoglobulin (presumably antibody) and complement to the arterioles and glomerular capillaries, where they can be visualized by immunofluorescent techniques. These immunoglobulin deposits on the vessel walls induce platelet aggregation in the glomerular capillaries leading to acute renal shutdown (figure 17.6). The possibility of damage to antibody-coated cells through antibody-dependent cellular cytotoxicity must also be considered.

Insidious and late rejection is associated with subendothelial deposits of immunoglobulin and C3 on the glomerular basement membranes which may sometimes be an expression of an underlying immune complex disorder (originally necessitating the transplant) or possibly of complex formation with soluble antigens derived from the grafted kidney.

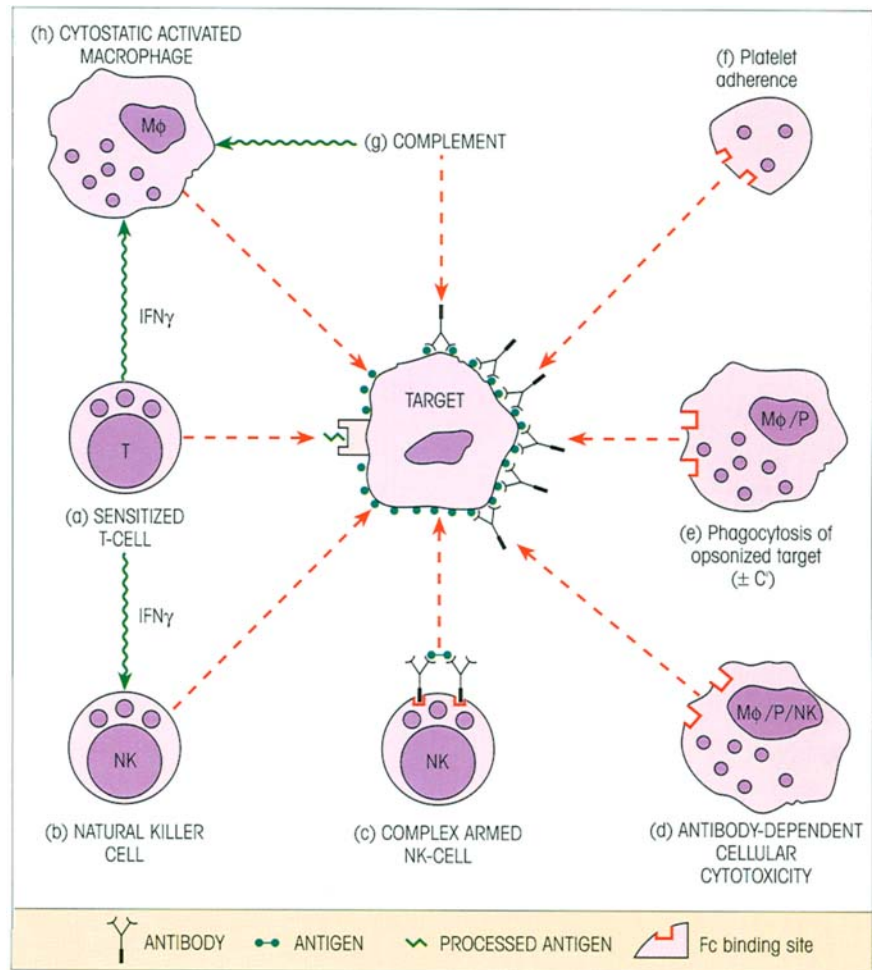
The complexity of the action and interaction of cellular and humoral factors in graft rejection is therefore considerable and an attempt to summarize the postulated mechanisms involved is presented in figure 17.7.

There are also circumstances when antibodies may actually *protect* a graft from destruction, a phenomenon termed *enhancement*.



Figure 17.6. Acute late rejection of human renal allograft showing platelet aggregation in a glomerular capillary induced by deposition of antibody on the vessel wall (electron micrograph). gbm, glomerular basement membrane; P, platelet. (Photograph courtesy of Professor K. Porter.)

Figure 17.7. Mechanisms of target cell destruction. M ϕ , macrophage; P, polymorph; NK, natural killer cell. (a) Direct killing by Tc cells and indirect tissue damage through release of cytokines from delayed-type hypersensitivity T-cells. (b) Killing by NK cells (see p. 18) enhanced by interferon. (c) Specific killing by immune complex-armed NK cell which recognizes the target through free antibody valencies in the complex. (d) Attack by antibody-dependent cellular cytotoxicity (in a–d the killing is extracellular). (e) Phagocytosis of target coated with antibody (heightened by bound C3b). (f) Sticking of platelets to antibody bound to the surface of graft vascular endothelium leading to formation of microthrombi. (g) Complement-mediated cytotoxicity. (h) Macrophages activated nonspecifically by agents such as BCG, endotoxin, poly-I:C, IFN γ and possibly C3b are cytostatic and sometimes cytotoxic for dividing tumor cells, perhaps through extracellular action of TNF and O $_2^-$ radicals generated at the cell surface (see p. 6).



THE PREVENTION OF GRAFT REJECTION

Matching tissue types on graft donor and recipient

Since MHC differences provoke the most vicious rejection of grafts, a prodigious amount of effort has gone into defining these antigen specificities, in an attempt to minimize rejection by matching graft and recipient in much the same way that individuals are cross-matched for blood transfusions (incidentally, the ABO group provides strong transplantation antigens).

HLA tissue typing

HLA alleles are now defined by their gene sequences and individuals can be typed by the polymerase chain reaction (PCR) using discriminating pairs of primers (figure 17.8). Molecules encoded by the class II *HLA-D* loci provoke CD4 T-cell responses, whereas *HLA-A*, *-B* and *-C* products are targets for alloreactive CD8 T-cells.

The polymorphism of the human HLA system

With so many alleles at each locus and so many loci in each individual (figure 17.9), it will readily be appreciated that this gives rise to an exceptional degree of polymorphism which is compounded further by the existence of multiple allotypic forms of the class III MHC complement components C2, C4a, C4b and factor B.

This remarkable polymorphism is of great potential value to the species, since the need for T-cells to recognize their own individual specificities provides a defense against microbial molecular mimicry in which a whole species might be put at risk by its inability to recognize as foreign an organism which generates MHC-peptide complexes similar to self. It is also possible that in some way the existence of a high degree of polymorphism helps to maintain the diversity of antigenic recognition within the lymphoid system of a given species and also ensures heterozygosity (hybrid vigor).

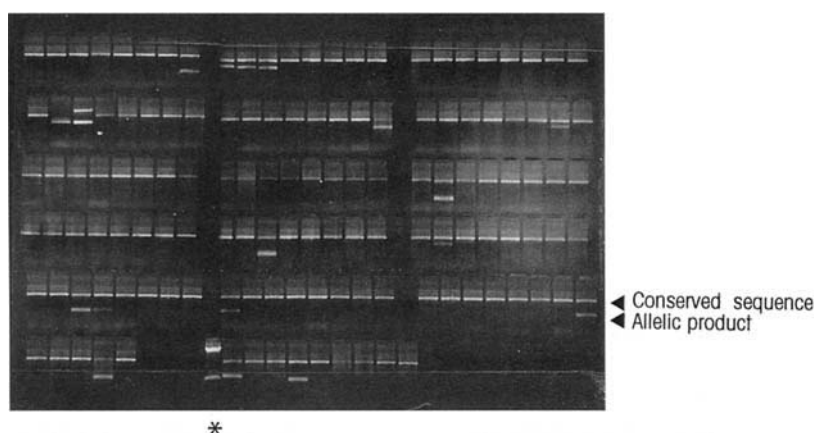


Figure 17.8. HLA tissue typing. Identification of HLA alleles depends upon polymerase chain reaction (PCR) amplification using sequence-specific primers. The photograph shows reactions of 96 primer pairs specific for each allele as listed in the protocol given by Bunce M. *et al.* (1995) *Tissue Antigens* 46, 355. A DNA sample to be typed was added to each PCR mix consisting of a pair of control primers (generating a 796 base pair product from a conserved DRB1 intron sequence) and primers to produce allele-specific products. After DNA amplification, the product was electrophoresed on ethidium bromide prestained gel and visualized by ultraviolet light. All

lanes contain a band corresponding to the control product, while positive reactions are those lanes containing an additional allele-specific band of predetermined size. This sample has the HLA type: A*2501,2601-04; B*1401/02,1801/02; Cw*0802,1203; DRB1*1501-05,0701; DRB4*0101-03; DRB5*0101/02/0201/02; DQB1*0201/02,0602. *This lane contains a DNA base pair ladder. The advantages of the method are correct assignment of homozygosity, higher resolution and improved accuracy. (Photograph kindly supplied by Dr D. Briggs.)

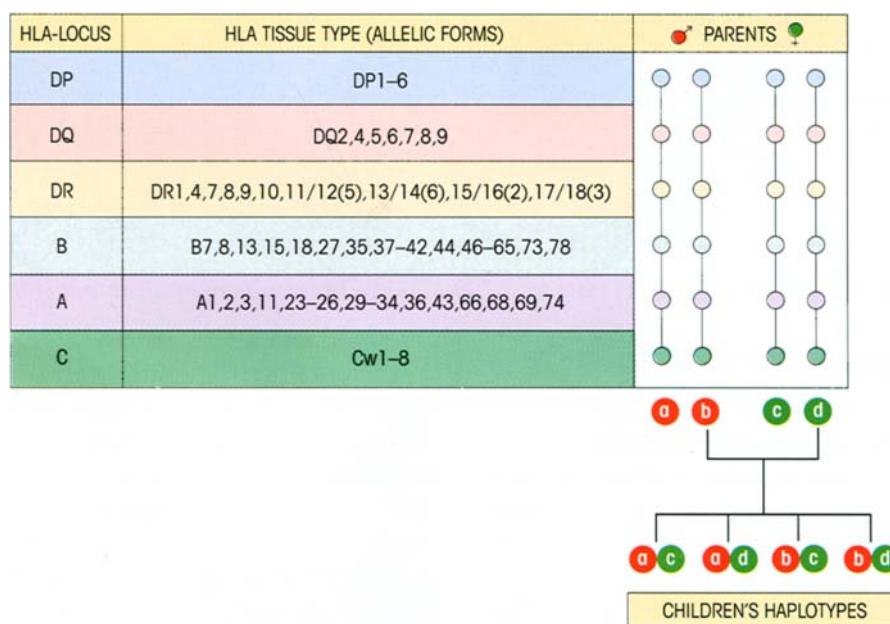


Figure 17.9. Polymorphic HLA specificities and their inheritance. The complex lies on chromosome 6, the DP locus being closest to the centromere. The numbers at the A and B loci do not overlap. The nomenclature for the DR alleles is set out in Bodmer J.G. *et al.* (1994) *Tissue Antigens* 44, 1. DR11/12(5) shows that the 'old pre-Bodmer' nomenclature DR5 can now be split by DNA analysis into two new specificities, DR11 and DR12. This list includes the most common alleles but is not complete! Since there are several possible alleles

at each locus, the probability of a random pair of subjects from the general population having identical HLA specificities is very low. However, there is a 1:4 chance that two siblings will be identical in this respect because each group of specificities on a single chromosome forms a haplotype which will be inherited *en bloc*, giving four possible combinations of paternal and maternal chromosomes. Parent and offspring can only be identical (1:2 chance) if the mother and father have one haplotype in common.

The value of matching tissue types

Improvements in operative techniques and the use of drugs such as cyclosporin A have greatly diminished the effects of mismatching HLA specificities on solid graft survival but, nevertheless, most transplanters favor a reasonable degree of matching (see figure 17.17). The consensus is that matching at the DR loci is of greater benefit than at the B loci, which in turn are of more relevance to graft survival than the A loci. In addition, the need for cross-matching to detect pre-sensitized recipients is now taken very seriously. Bone marrow grafts, however, require a high degree of compatibility and the greater accuracy of DNA typing methods can be most helpful in this respect.

Because of the many thousands of different HLA phenotypes possible (figure 17.9), it is usual to work with a large pool of potential recipients on a continental basis (Eurotransplant), so that when graft material becomes available the best possible match can be made. The position will be improved when the pool of

available organs can be increased through the development of long-term tissue storage banks, but techniques are not good enough for this at present, except in the case of bone marrow cells which can be kept viable even after freezing and thawing. With a paired organ such as the kidney, living donors may be used; siblings provide the best chance of a good match (cf. figure 17.9). However, the use of living donors poses difficult ethical problems and there has been encouraging progress in the use of cadaver material. There is active interest in the possibility of using animal organs (see below) or mechanical substitutes, while some are even trying to prevent the disease in the first place!

Agents producing general immunosuppression

Graft rejection can be held at bay by the use of agents which nonspecifically interfere with the induction or expression of the immune response (figure 17.10). Because these agents are nonspecific, patients on immunosuppressive therapy tend to be susceptible

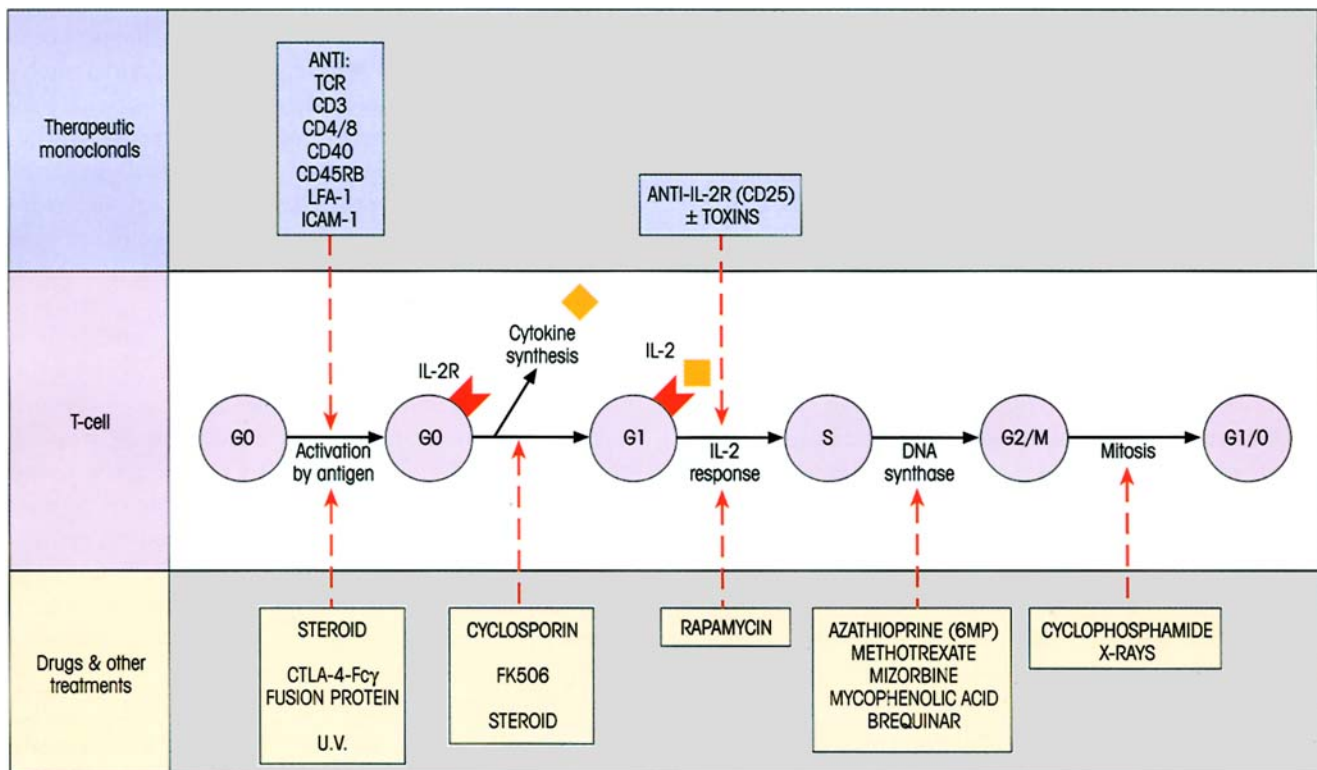


Figure 17.10. Immunosuppressive agents used to control graft rejection. The new name for FK506 is tacrolimus. Mycophenolic acid, a purine analog produced by metabolism of mycophenolate mofetil, is a powerful new immunosuppressant undergoing early trials which inhibits proliferation but also suppresses expression of CD25, -71, -154 (CD40L) and CD28. Another potent drug is deoxyspergualin which interferes with lymphocyte function in an, as yet, ill-

understood manner. Leflunomide effectively blocks T-dependent and -independent antibody synthesis *in vivo* by inhibiting dihydroorotate dehydrogenase, an enzyme required for *de novo* synthesis of uridine 5'-PO₄. Simultaneous treatment with agents acting at sequential stages in development of the rejection response would be expected to lead to strong synergy and this is clearly seen with cyclosporin A and rapamycin.

to infections; they are also more prone to develop lymphoreticular cancers, particularly those of viral etiology.

Targeting lymphoid populations

Anti-CD3 monoclonals are in widespread use as anti-T-cell reagents to reverse acute graft rejection. The potential therapeutic benefits though have been constrained by their immunogenicity and their propensity to activate a severe cytokine release syndrome involving complex 'flu-like' symptoms. These problems can be circumvented by 'humanizing' the antibody (see p. 123) and by mutating position 297 to prevent glycosylation and consequently binding to Fc receptors and to complement. An IgM monoclonal to conserved regions on the TCR now available seems to provoke fewer untoward reactions than conventional anti-CD3.

The IL-2 receptor α chain (CD25), expressed by activated but not resting T-cells, represents another exploitable target. A humanized version of a murine monoclonal anti-IL-2R α , dubbed daclizumab (a name seemingly derived from a Harry Potter story), had a longer half-life and of course reduced immunogenicity. When used to supplement standard baseline immunosuppression consisting of cyclosporin plus corticosteroids, acute rejection episodes at 6 months in recipients of cadaveric kidney grafts occurred in only 28% as compared with 47% in the placebo controls. The results were sufficiently encouraging for the FDA to provide market clearance for the use of daclizumab in the prevention of acute kidney transplant rejection. Following the relative ease of engraftment of allogeneic bone marrow in patients with a deficiency of the 'adhesin' molecule LFA-1 (see p. 306), attention is now turning to the use of anti-LFA-1 as an immunosuppressant for such grafts.

Immunosuppressive drugs

The development of an immunological response requires the active proliferation of a relatively small number of antigen-sensitive lymphocytes to give a population of sensitized cells large enough to be effective. Many of the immunosuppressive drugs now employed were first used in cancer chemotherapy because of their toxicity to dividing cells. Aside from the complications of blanket immunosuppression mentioned above, these antimitotic drugs are especially toxic for cells of the bone marrow and small intestine and must therefore be used with great care.

A commonly used drug in this field is **azathioprine**

which has a preferential effect on T-cell-mediated reactions. It is broken down in the body first to 6-mercaptopurine and then converted to the active agent, the ribotide. Because of the similarity in shape, this competes with inosinic acid for enzymes concerned with the synthesis of guanylic and adenylic acids; it also inhibits the synthesis of 5-phosphoribosylamine, a precursor of inosinic acid, by a feedback mechanism. The net result is inhibition of nucleic acid synthesis. Another drug, **methotrexate**, through its action as a folic acid antagonist also inhibits synthesis of nucleic acid. The N-mustard derivative **cyclophosphamide** probably attacks DNA by alkylation and cross-linking, so preventing correct duplication during cell division. These agents appear to exert their damaging effects on cells during mitosis and, for this reason, are most powerful when administered after presentation of antigen at a time when the antigen-sensitive cells are dividing.

An exciting group of fungal metabolites is having a dramatic effect in human transplantation and in the therapy of immunological disorders through its ability to target T-cells. **Cyclosporin A** (CsA), a neutral hydrophobic cyclical peptide containing 11 amino acids which is extremely insoluble, selectively blocks the transcription of IL-2 in activated T-cells. Resting cells which carry the vital memory for immunity to microbial infections are spared and there is little toxicity for dividing cells in gut and bone marrow. Some studies also point to an 'exquisite' sensitivity of dendritic APCs to the drug. Another T-cell-specific immunosuppressive drug, **FK506**, isolated from a species of *Streptomyces*, also blocks cytokine production. The latest addition to the stable is **rapamycin**, a product of the fungus *Streptomyces hygroscopicus*, which is a macrolide like FK506, but in contrast acts to block signals induced by combination of IL-2 with its receptor.

We now have greater insight into the mode of action of these drugs (figure 17.11). Both CsA and FK506 complex with different specific binding proteins termed **immunophilins** (cyclophilin and FK-binding protein respectively), which for obscure reasons have peptidyl-prolyl isomerase activity; these complexes then interact with and inhibit the calcium- and calmodulin-dependent phosphatase, calcineurin A, which is responsible for producing the transcription factors for IL-2, apoptosis and exocytosis in activated T-cells. Transcription of IL-10 is spared suggesting that switching from Th1 to Th2 responses may be a consequence. Although rapamycin also binds to the FK-binding protein, the complex has a quite different biological activity in that it blocks the IL-2-induced activation of the p70 S6 kinase which phosphorylates ribosomal S6 prior to cell proliferation.

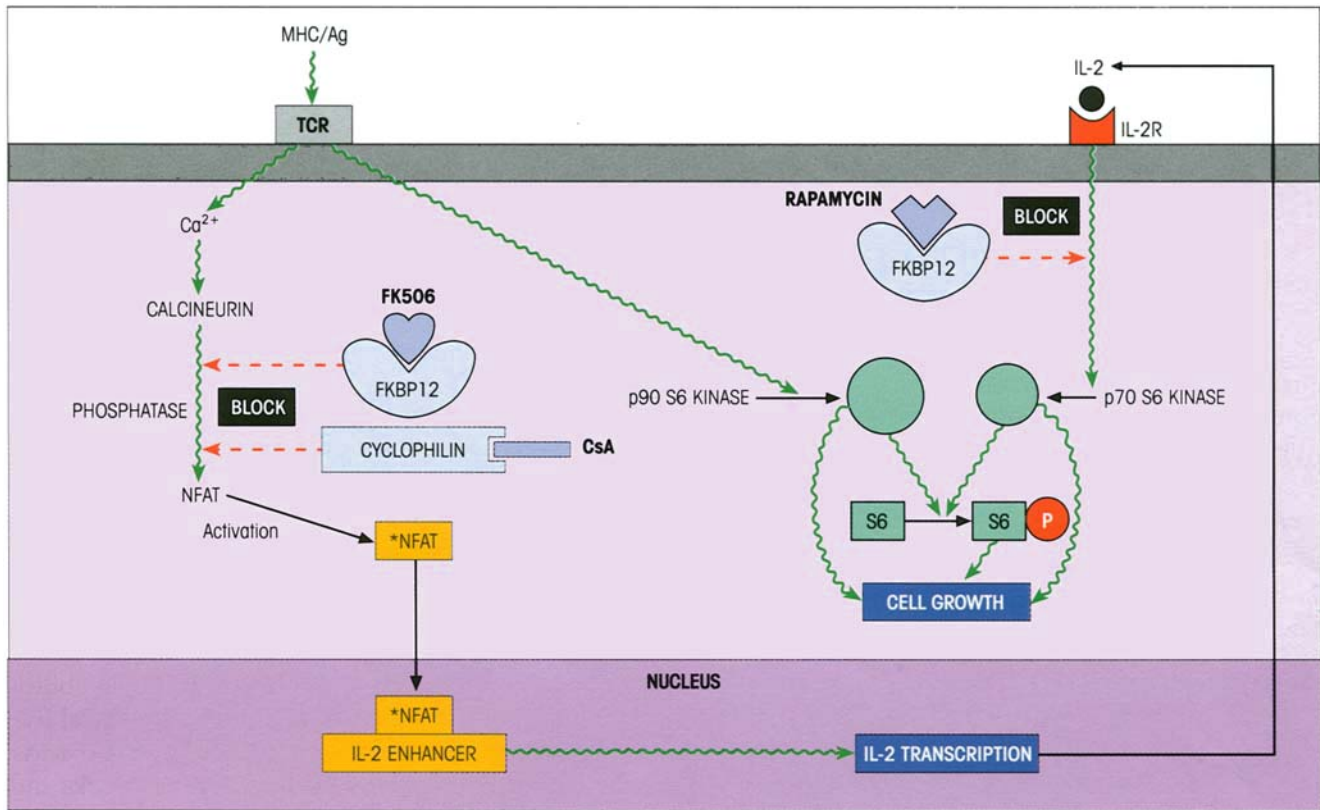


Figure 17.11. The mode of action of cyclosporin, FK506 and rapamycin. The complexes of CsA with cyclophilin and of FK506 with FKBP12 (one of a family of FK506-binding proteins) bind to and inactivate the phosphatase calcineurin responsible for activating nuclear factor of activated T-cells (NFAT) (and possibly OAP and Oct-1; cf. p. 168), which is a transcription factor for IL-2 synthesis. Transfection with calcineurin decreases the inhibitory powers of CsA and FK506. On binding to cyclophilin, CsA undergoes an awesome conformational change enabling it to exteriorize hydrophobic side-

chains to form a patch which can bind calcineurin, rather like double-sided tape. Aminodextran derivatized with approximately 10 molecules of CsA agglutinated T-cells and, although it cannot penetrate the lymphocytes, it inhibited phorbol ester-induced IL-2 synthesis, suggesting a reaction with cyclophilin on the cell membrane rather than in the cytoplasm. The rapamycin-FKBP12 complex blocks the activation of p70 S6 kinase by transduced IL-2 signals, thereby inhibiting cell growth.

Cyclosporin now has a proven place as first-line therapy in the prophylaxis and treatment of transplant rejection. Figure 17.12 gives an example of its use in kidney transplantation, but it has also been evaluated in a wide range of disorders where T-cell-mediated hypersensitivity reactions are suspected. Indeed, the benefits of cyclosporin in diseases such as idiopathic nephrotic syndrome, type 1 insulin-dependent diabetes, Behçet's syndrome, active Crohn's disease, aplastic anemia, severe corticosteroid-dependent asthma and psoriasis have been interpreted to suggest or confirm a pathogenic role for the immune system. However, effects not only on Langerhans' dendritic cells but also on the proliferation of normal and transformed keratinocytes *in vitro* may contribute to the favorable outcome in psoriasis. A rapid onset of benefit and relapse when treatment is stopped are common features of cyclosporin therapy.

There are, of course, side-effects. It has to be used at

doses below those causing renal fibrosis due to stimulation of TGF β production by several cell types. There is also some cause for concern that cyclosporin may make patients susceptible to EB virus-induced lymphomas since the drug inhibits T-cells which control EB virus transformation of B-cells *in vitro*; however, the latest results suggest that the incidence of lymphoma is relatively low in comparison with that reported for allografted patients on conventional immunosuppressive therapy. Although not itself carcinogenic, cyclosporin can promote tumor progression, probably through its effect on TGF β .

FK506 is greatly superior to CsA on a molar basis *in vitro* but is not substantially more effective when used in kidney grafting; however, its tropism for liver could be exploited. Because they act at different stages in the activation of the T-cell, CsA and rapamycin show a most impressive degree of synergy which allows the two drugs to be used at considerably lower dose levels

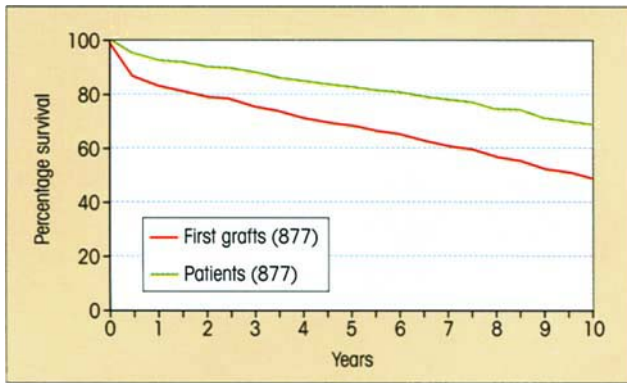


Figure 17.12. Actuarial survival of primary cadaveric kidney grafts in 877 patients treated at the Oxford Transplant Centre with triple therapy. **Cyclosporin:** 10 mg/kg/day orally given as a single dose (first dose given before surgery) and reduced according to whole-blood trough levels (200–400 ng/ml in first 3 months and then maintained at 100–200 ng/ml). **Azathioprine:** 100 mg/day orally; reduced if leukocyte count < 5000. **Prednisolone:** 20 mg/day given in divided doses orally and reduced to a maintenance dose of 10 mg/day by 6 months. The initial dose is reduced to 15 mg/day in patients of less than 60 kg body weight. In patients with stable renal function at 1 year, prednisolone is discontinued over a period of some months. (Data kindly provided by Professor Peter J. Morris.)

with correspondingly less likelihood of side-effects (figure 17.10). Another possible synergistic partner for cyclosporin is fludarabine which, unlike cyclosporin, blocks signaling by STAT-1, an intracellular intermediate activated by interferons and important for cell-mediated immunity. Combination therapies may also include azodicarbonamide which blocks the calcium flux associated with CD3 activation and the drugs mycophenolic acid and leflunomide which limit the availability of DNA precursors.

Steroids such as prednisolone intervene at many points in the immune response, affecting lymphocyte recirculation and the generation of cytotoxic effector cells, for example; in addition, their outstanding anti-inflammatory potency rests on features such as inhibition of neutrophil adherence to vascular endothelium in an inflammatory area and suppression of monocyte/macrophage functions such as microbicidal activity and response to cytokines. Corticosteroids form complexes with intracellular receptors which then bind to regulatory genes and block transcription of TNF, IFN γ , IL-1, -2, -3, -6 and MHC class II, i.e. they block expression of cytokines from both lymphocytes and macrophages, whereas cyclosporin has its main action on the former.

In parentheses, the immunophilins may be involved in other cellular functions, such as the regulation of the *N*-methyl-D-aspartate subtype of neural glutamate receptors and the augmentation of neuronal process ex-

tension by growth-associated protein-43, leading to the propositions that FK506 could be clinically employed in stroke patients and in the treatment of nerve degeneration. A close association of FKBP12 with the surface receptor for transforming growth factor- β (TGF β) has also been revealed. With good fortune these drugs will take us down some very unexpected and intriguing pathways.

Strategies for antigen-specific depression of allograft reactivity

If the disadvantages of blanket immunosuppression are to be avoided, we must aim at knocking out only the reactivity of the host to the antigens of the graft, leaving the remainder of the immunological apparatus intact—in other words, the induction of **antigen-specific tolerance**.

It turns out that bone marrow represents a privileged source of tolerogenic alloantigens, and the production of stable lymphohematopoietic mixed chimerism by bone marrow engraftment is proving to be a potent means of inducing robust specific transplantation tolerance to solid organs across major MHC mismatches. However, successful allogeneic bone marrow transplantation in immunocompetent adults normally requires cytoablative treatment of recipients with irradiation or cytotoxic drugs and this has tended to restrict its use to malignant conditions. A most encouraging recent study has now shown the feasibility of inducing long-lasting tolerance not only to bone marrow cells but also to fully MHC-mismatched skin grafts in naive recipients receiving high-dose bone marrow transplantation and costimulatory blockade by single injections of monoclonal anti-CD154 (CD40L) plus a CTLA-4-Ig fusion protein (figure 17.13). A persistent hematopoietic macrochimerism is achieved with a significant proportion of donor-type lymphocytes in the thymus indicating intrathymic deletion of donor-reactive T-cells.

While this protocol permits long-term engraftment of bone marrow and solid organs, it seems that direct blockade with just anti-CD154 and CTLA-4-Ig is sufficient to induce tolerance to solid organ grafts. Stimulation of alloreactive T-cells by the graft in the presence of costimulatory blockade leads to apoptosis, a process promoted by rapamycin which improves the tolerant state. Bcl-X_L (cf. figure 11.5, p. 203) prevents both T-cell apoptosis and tolerance induction by this treatment revealing the importance of apoptotic T-cell deletion for the establishment of antigen-specific unresponsiveness. In a further twist to the tale, the apoptotic T-cells 'reach from beyond the grave' by producing IL-10, so

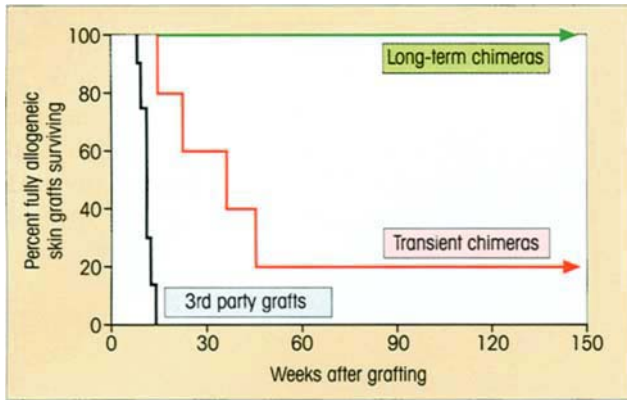


Figure 17.13. Induction of tolerance and macrochimerism by fully allogeneic bone marrow transplantation plus costimulatory blockade. B6 mice received bone marrow cells from the fully allogeneic B10.A strain with injections of anti-CD154 (CD40L) and the CTLA-4-Ig fusion protein which blocks B7-CD28 interactions (CTLA is a T-cell receptor for B7 which downregulates T-cell activation; cf. pp. 165 and 170). Eight mice showing long-term persistence of multilineage donor cells (macrochimerism) were fully tolerant to B10.A skin grafts. Five mice with transient chimerism showed moderate prolongation of skin graft survival relative to unrelated 3rd party grafts. (Data taken from Wekerle T. *et al.* (2000) *Nature Medicine* 6, 464, with permission.)

that their phagocytosis along with antigen leads to the presentation of the antigen in a tolerogenic form which maintains tolerance through the production of immunoregulatory cells.

Despite the role of the *mature* dendritic cell as the champion stimulator of resting T-cells, the dendritic cell *precursors* may present antigen in the absence of B7 costimulators and, by mechanisms echoing those described above in the costimulatory blockade experiments, would appear to have a powerful potential for tolerance induction. This concept is of particular relevance to the specific unresponsiveness generated by grafts of liver, which being a hematopoietic organ, continually exports large numbers of these immature dendritic cells.

Nondepleting anti-CD4 and -8 monoclonals, by depriving T-cells of fully activating signals, can render them anergic when they engage antigen through their specific receptors. These anergic cells can induce unresponsiveness in newly recruited T-cells ('infectious tolerance', p. 233) and so establish specific and indefinite acceptance of mouse skin grafts across class I or multiple minor transplantation antigen barriers (figure 17.14). It should be noted that skin allografts provide the most difficult challenge for tolerance induction, and transplants of organs such as the heart, which are less fastidious than skin, require less aggressive immunotherapy.

Given the wide variety of different peptide epitopes presented by the graft MHC, full-frontal attack on the

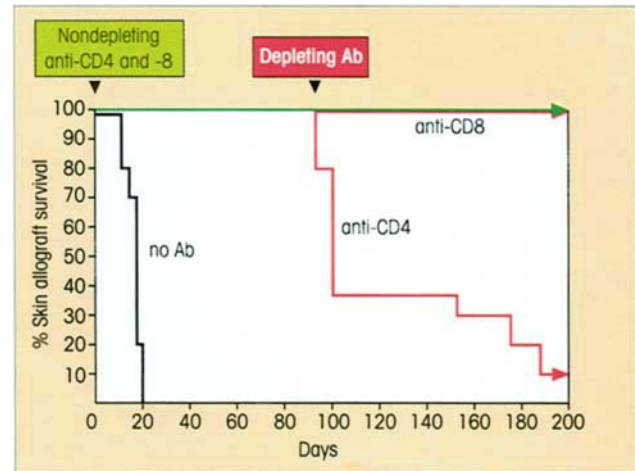


Figure 17.14. Induction of allograft tolerance by nondepleting anti-CD4 plus anti-CD8. Tolerance to skin grafts from donors with multiple minor transplantation antigen mismatches was achieved by concurrent injection of IgG2a monoclonal antibodies to CD4 and CD8 which do not induce cell depletion (green arrow). The maintenance of tolerance depends upon the continued presence of antigen which enables the unresponsive cells to interact with newly arising immunocompetent cells on the surface of the same antigen-presenting cells and render them unresponsive through an infectious tolerance mechanism (cf. figures 12.9 and 20.24). Whether the mechanism involves 'infectious anergy', direct suppression, 'immune deviation' of potentially aggressive Th1 cells by suppressor Th2, down-regulation of the antigen-presenting cell, or all four in various combinations, is unresolved. Loss of tolerance on depletion of CD4 but not CD8 cells (red arrows) shows that active tolerance is maintained by the CD4 subset. (Figure synthesized from data kindly provided by Dr S.P. Cobbold and Professor H. Waldmann.)

alloreactive T-cells by administration of tolerogenic peptides represents quite a challenge, and the strategy of using costimulatory blockade with the antigens being provided by the graft itself looks to be a more promising route.

IS XENOGRAFTING A PRACTICAL PROPOSITION?

Because the supply of donor human organs for transplantation lags seriously behind the demand, a widespread interest in the feasibility of using animal organs is emerging. Pigs are more favored than primates as donors both on the grounds of ethical acceptability and the hazards of zoonoses. The first hurdle to be overcome is **hyperacute rejection** due to xenoreactive natural antibodies in the host. Humans lack α -1,3-galactosyl transferase and galactose α -1,3-galactose and therefore develop antibodies to this epitope which is present on many common bacteria and expressed abundantly on the xenogeneic vascular endothelium. The natural antibodies bind to the endothelium and activate complement in the absence of regulators of the

Genetic manipulation of pig	Graft	+ Human natural anti-Gal + C'	Hyperacute rejection
-		→	++
Make transgenic for human DAF or CD59		→	-
Transfect cells with α -1,2-fucosyl transferase		→	-

Figure 17.15. Strategies for avoiding complement-mediated hyperacute rejection of a xenograft caused by reaction of natural anti-galactose antibodies with Gal α -1,3-Gal on the surface of the pig graft cells. Heart xenografts from transgenic pigs expressing the human complement regulatory proteins decay accelerating factor (DAF) or CD59 functioned for prolonged periods in baboons. Transfection of

pig cells with α -1,2-fucosyl transferase converted the terminal sugars into the blood group H and rendered the cells resistant to lysis by the anti-galactose. Other strategies involve transfection with genes encoding an α -galactosidase or intracellular recombinant scFv reacting with α -1,3-galactosyl transferase.

human complement system, such as decay accelerating factor, CD59 and MCP (cf. figure 15.2), precipitating the hyperacute rejection phenomenon. Novel genetic engineering strategies for the solution of this problem are outlined in figure 17.15.

The next crisis is **acute vascular rejection** occurring within 6 days as antibodies produced by Th2 cells react with the xenoantigens on donor epithelium. In contrast with the response to allografts, IL-12 and IFN γ actually inhibit acute vascular rejection and, over the long term, IFN γ may protect the graft by promoting the formation of NO $\cdot$ which prevents constriction of blood vessels. Brequinar sodium (figure 17.10), an inhibitor of pyrimidine biosynthesis and suppressor of both B- and T-cell-mediated responses, has been evaluated for efficacy, but induction of tolerance would clearly be more desirable although tricky.

Even when the immunological problems are overcome, it remains to be seen whether the xenograft will be compatible with human life over a prolonged period, latent pig viruses being an obvious worry.

TISSUE ENGINEERING

The ideal would be to create the tissue for a **graft entirely from cells of the recipient**, i.e. an autograft, which would eliminate the need for immunosuppression. Science fiction? Probably not. We will steadily accumulate the knowledge concerning the various growth factors required to guide relatively undifferentiated stem cells into the desired mature form under culture conditions *in vitro*. In many instances, it is proving possible to isolate stem cells from various adult or-

gans, but the most exciting development has come from cloning 'Dolly' the sheep. This has revealed the feasibility of taking the nucleus from a differentiated adult cell and reprogramming it into an embryonic state responsive to early growth factors, which can be selected to determine its future destiny as an islet, brain or liver cell or what have you (figure 17.16).

CLINICAL EXPERIENCE IN GRAFTING

Privileged sites

Corneal grafts survive without the need for immunosuppression. Because they are avascular they tend not to sensitize the recipient. This privileged protection is boosted by the local production of immunosuppressive factors such as TGF β , IL-1Ra, limited expression of MHC and the strategic presence of FasL which can induce apoptosis in infiltrating lymphocytes. Nonetheless, they do become cloudy if the individual has been *presensitized*. Grafts of **cartilage** are successful in the same way but an additional factor is the protection afforded the chondrocytes by the matrix. With bone and artery it doesn't really matter if the grafts die because they can still provide a framework for host cells to colonize.

Kidney grafts

Thousands of kidneys have been transplanted and with improvement in patient management there is a high survival rate. In the long term (5 years or more), the desirability of reasonable matching at

Figure 17.16. Anticipated production of autologous grafts by tissue engineering. Undifferentiated cells are obtained directly from the patient either as stem cells (if available) or by transplanting the nuclei of mature cells into enucleated embryonic cells. They are cultured in a biodegradable matrix with appropriate growth factors to provide a tissue populated with differentiated cells which can function as an autologous graft.

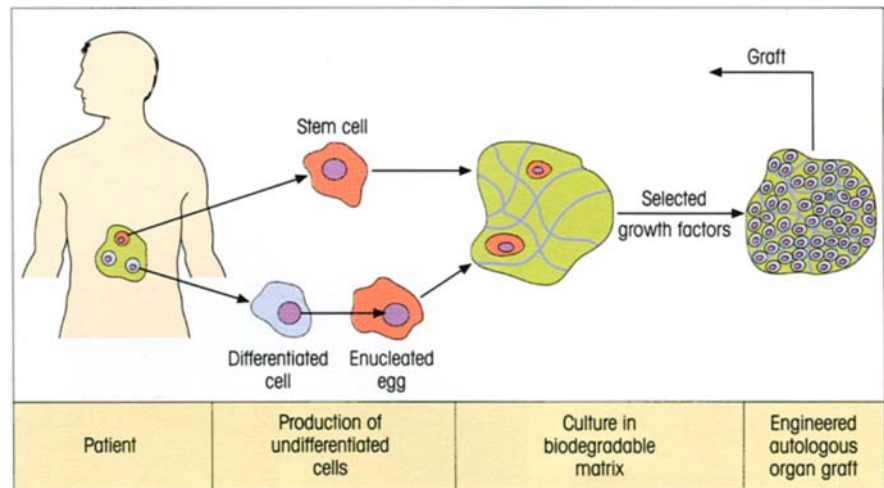
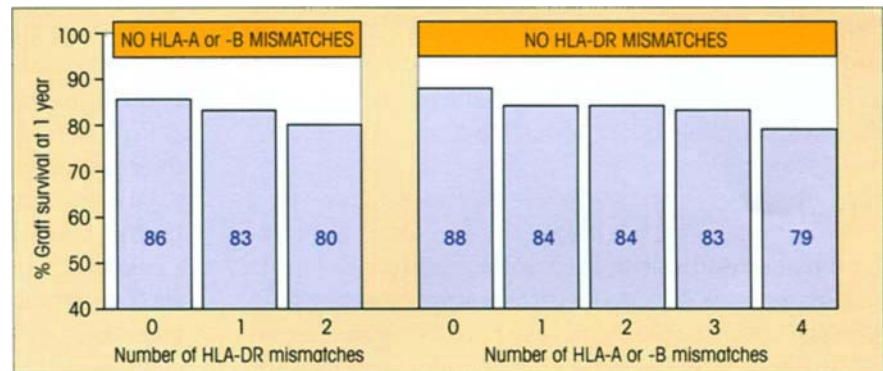


Figure 17.17. First cadaveric kidney graft survival in Europe for the period January 1993 to December 1997 ($n=12584$) on the basis of mismatches for HLA-A, -B and -DR. There is a significant influence of matching, $p<0.001$, for both sets of data. (Data kindly supplied by Drs Guido Persijn and Jacqueline Smits of the Eurotransplant International Foundation.)



the HLA-A, -B and -D loci becomes apparent (figure 17.17).

Patients are partially immunosuppressed at the time of transplantation because uremia causes a degree of immunological anergy. The combination of azathioprine and prednisolone was commonly employed in the long-term management of kidney grafts, but is now supplemented by cyclosporin A in the so-called **triple therapy** (figure 17.12). One hopes that the synergy between cyclosporin and rapamycin, and possibly other immunosuppressants, will lead to the emergence of powerful new therapeutic regimens. If kidney function is poor during a rejection crisis, renal dialysis can be used. As mentioned above, there is active interest in the possibility of xenografting. When transplantation is performed because of immune complex-induced glomerulonephritis, the immunosuppressive treatment used may help to prevent a similar lesion developing in the grafted kidney. Patients with glomerular basement membrane antibodies (e.g. Goodpasture's syndrome) are likely to destroy their renal transplants unless first treated with plasmapheresis and immunosuppressive drugs.

Heart transplants

The overall 1-year survival figure for heart transplants has moved up to near the 80% mark (figure 17.18), helped considerably by the introduction of cyclosporin A therapy. Its nephrotoxicity is a drawback, although the analog cyclosporin G is better in this respect, and synergic combination therapy with rapamycin should reduce this side-effect markedly. Full HLA matching is of course not practical, but single DR mismatches gave 90% survival at 3 years compared with a figure of 65% for two DR mismatches. Aside from the rejection problem, it is likely that the number of patients who would benefit from cardiac replacement is much greater than the number dying with adequately healthy hearts. More attention will have to be given to the possibility of xenogeneic grafts and mechanical substitutes.

One interesting nugget. The immune response to adenoviral vectors can induce inflammation and loss of gene expression in transfected tissues, except for cardiac isografts which remain free of inflammation. This suggests a novel way of introducing an immuno-

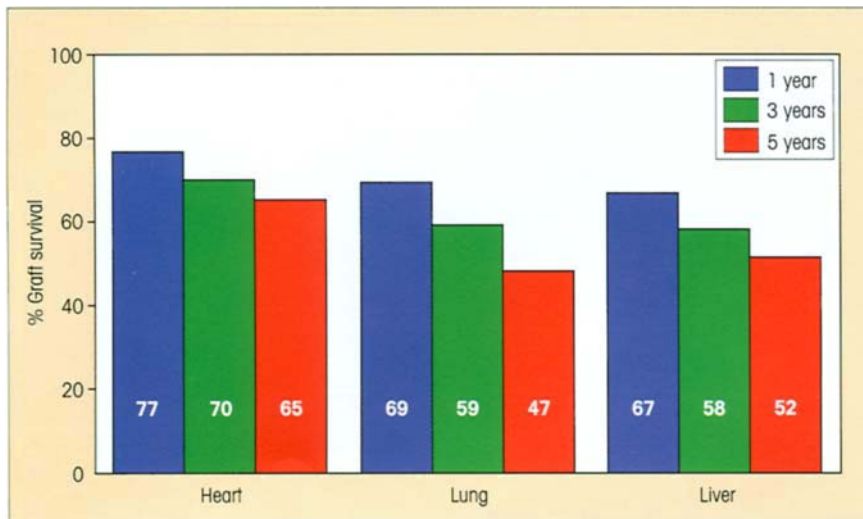


Figure 17.18. Graft survival rates for first heart, liver and lung transplantations in Eurotransplant performed between January 1992 and December 1993. (Data kindly supplied by Drs Guido Persijn and Jacqueline Smits, Eurotransplant International Foundation.)

suppressant such as TGF β , which can be produced in high concentrations within the milieu of a cardiac graft without invoking systemic side-effects.

Liver transplants

Survival rates for orthotopic liver grafts are not quite as high as those achieved with heart transplants (figure 17.18). The hepatotrophic capacity of FK506 is an added bonus which makes it the preferred drug for liver transplantation. Rejection crises are dealt with by high-dose steroids and, if this proves ineffective, anti-lymphocyte globulin. The use of a totally synthetic colloidal hydroxyethyl starch solution, containing lactobionate as a substitute for chloride, allows livers to be preserved for 24 hours or more and has revolutionized the logistics of liver transplantation. To improve the prognosis of patients with primary hepatic or bile duct malignancies, which were considered to be inoperable, transplantation of organ clusters with liver as the central organ has been designed, e.g. liver and pancreas, or liver, pancreas, stomach and small bowel or even colon. Nonetheless, the outcome is not very favorable in that up to three-quarters of the patients transplanted for hepatic cancer have recurrence of their tumor within 1 year. For the future we must look forward to the creation of autologous liver from adult cells when tissue engineering techniques have been developed sufficiently.

Experience with liver grafting between pigs revealed an unexpected finding. Many of the animals retained the grafted organs in a healthy state for many months without any form of immunosuppression and enjoyed a state of unresponsiveness to grafts of skin or kidney from the same donor. True tolerance is induced

by the donor-type intrahepatic hematopoietic stem cells and immature dendritic cells (see above) and possibly also by the liver parenchyma itself, known to produce copious amounts of soluble MHC class I.

Work is in progress on the transfer of isolated hepatocytes attached to collagen-coated microcarriers injected i.p. for the correction of isolated deficiencies such as albumin synthesis. This attractive approach has much wider application as a general vehicle for gene therapy.

Bone marrow grafting

Patients with certain immunodeficiency disorders and aplastic anemia are obvious candidates for treatment with **bone marrow stem cells** with their potential to differentiate into all the formed elements of the blood; so, too, are acute leukemia patients treated radically with intensive chemotherapy and possibly whole-body irradiation in attempts to eradicate the neoplastic cells, as will be discussed in the next chapter.

Bone marrow contains not only hematopoietic but also mesenchymal stem cells which can give rise to cartilage, tendons and bone; after expansion in culture by a factor of 5–10 times, they provide an excellent treatment for children with osteogenesis imperfecta, a genetic disorder in which the osteoblasts produce defective type I collagen with resulting osteopenia and severe bony deformities. Good results are being obtained with stem cell transplantation *in utero* for inherited blood disorders using populations from paternal bone marrow enriched for the stem cell marker, CD34. From the practical standpoint, it has been recognized that cord blood contains sufficient hematopoietic stem cells for bone marrow replacement, but what is even

more convenient is to enhance the number of CD34-positive progenitor cells in relatively small volumes of peripheral blood by recombinant stem cell factor, IL-1 β , IL-3, IL-6 and erythropoietin in the presence of stromal cells or fibronectin monolayers *ex vivo* prior to transplantation. It is encouraging also that both colony-stimulating factors G-CSF and GM-CSF greatly accelerate the return of myeloid cells after lethal doses of myelotoxic agents preceding autologous bone marrow transplantation; there are fewer infections, less use of i.v. antibodies and earlier discharge from hospital. They also accelerate the engraftment of allogeneic bone marrow without exacerbating g.v.h. disease.

Graft-vs-host disease is a major problem in bone marrow grafting

G.v.h. disease resulting from the recognition of recipient antigens by allogeneic T-cells in the bone marrow inoculum represents a serious, sometimes fatal, complication, and the incidence of g.v.h. disease is reduced if T-cells in the grafted marrow are first depleted with a cytotoxic cocktail of anti-T-cell monoclonals. Unexpectedly, complete purging of the T-cells leads to an alarmingly high incidence of graft failure, apparently due to the loss of relatively rare facilitating cells expressing TCR, CD3, CD8 and a previously unknown 33 kDa cell surface glycoprotein, FCp33. How these and other facilitating cells, such as CD8⁺ dendritic cells, work is, to put it mildly, unknown.

It is fondly hoped that successful engraftment and avoidance of g.v.h. reactions will be achieved in the clinic—in the not too distant future(?)—by strategies such as those involving allogeneic marrow transplantation under costimulatory blockade without cytoablative treatment of graft or recipient, as described earlier (figure 17.13). Until then, successful results are more likely with highly compatible donors, particularly if fatal g.v.h. reactions are to be avoided, and here siblings offer the best chance of finding a matched donor (figure 17.9). Undoubtedly, non-HLA minor transplantation antigens are important and are more difficult to match. Acute g.v.h. disease occurring within the first 100 days following infusion of allogeneic marrow primarily affects the skin, liver and gastrointestinal tract. Antibodies to TNF or IL-1R block mortality. Current therapy uses cyclosporin with prednisolone, but inclusion of methotrexate in this regimen is said to improve efficacy. Chronic g.v.h. disease (i.e. later than 100 days) has a relatively good prognosis if limited to skin and liver, but if multiple organs are involved, clinically resembling progressive systemic

sclerosis, the outcome is poor. Patients are treated with cyclosporin and prednisolone, but recently thalidomide has been found to be very helpful because of its anti-TNF effect. The pathogenesis is not straightforward and chronic disease could arise by a curious mechanism involving the sneaking through of autoreactive T-cells, which fail to be deleted in the thymus possibly because of thymic damage due to pre- or post-transplant therapy, e.g. associated viral infection, irradiation or even cyclosporin itself. It may be significant that cyclosporin inhibits the programmed cell death of immature thymocytes which occurs on activation by anti-CD3, and it is known that an autologous 'g.v.h.' reaction supervenes on termination of prolonged cyclosporin administration to young irradiated rats which had received syngeneic bone marrow.

Other organs

It is to be expected that improvement in techniques of control of the rejection process will encourage transplantation in several other areas—not in cases of endocrine disorders such as Hashimoto's disease (cf. p. 397) where exogenous replacement therapy is convenient, but, for example, in diabetes where the number of transplants recorded is rising rapidly and the current success rate is around 40%, although major problems still are inadequate output of insulin by the grafted islets and an adequate supply of tissue. Considerable advances have been made in very substantially expanding the number of islet cells which can be derived from pancreatic duct precursors, while monkey islets may not prove to be a bad bet in that they have low surface MHC and are nonvascularized, so avoiding the presentation of the dangerous GalGal epitope (cf. p. 361) which is enriched on vascular endothelium in some marked species. *In vivo* transfection of hepatocytes with an adenoviral vector encoding the homeoprotein PDX-1, which is a glucose-dependent transactivator of the insulin gene, induced them to make insulin in diabetic mice—very impressive, lots of potential, but early days yet. The 5-year survival rate of 47% for lung transplants is improving but is still less than satisfactory. One also looks forward to the day when the successful transplantation of skin for lethal burns becomes more commonplace.

Reports are coming in of an experimental foray into the grafting of **neural tissues**. Mutant mice with degenerate cerebellar Purkinje cells, which mimic the human condition cerebellar ataxia, can be restored by engraftment of donor cerebellar cells at the appropriate sites; these express insulin-like growth factor-1 (IGF-1), migrate to form a layer in lieu of the missing cells, induce

sprouting in host neurons and become synaptically integrated. Clinical trials with transplantation of human embryonic dopamine neurons to reverse the neurological deficit in Parkinson's disease have been severely hampered by the excessive death of the grafted cells. The hypothesis that this was due to oxidative stress led to a study showing that grafted neurons from transgenic mice overexpressing Cu/Zn superoxide dismutase (cf. p. 6) had a greatly increased survival rate. Another candidate is Huntington's chorea which has been tracked down to a mutation producing a protein with an expanded polymorphic glutamine repeat, toxic to spiny striatal neurons. Embryonic striatal allografts, producing the nontoxic wild-type protein, survive for a prolonged period, become integrated into the host brain and induce functional alleviation of motor deficits, all in the absence of immunosuppression. To date, the need to use embryonic tissue has been a bugbear, but the recent identification of neurogenic precursor cells from the adult human hippocampus may give a glimpse of the Promised Land.

Cryopreservation of sperm is a successful strategy in the management of adult cancer sufferers to protect the sperm from mutagenic cancer treatment. This is not available to prepubertal boys, but an alternative for them is cryopreservation of their spermatogonial stem cells for reintroduction post-treatment, since the Sertoli cells which support differentiation into mature spermatozoa will function normally. As knowledge from the human genome sequence accumulates, there is a potential for identifying and correcting genetic defects in the spermatogonia before their reintroduction, but ethical committees fight shy of this sort of 'Frankenstein' tinkering. More acceptably, in cases of male infertility due to dysfunctional Sertoli cells, it should be possible to develop mature spermatids by culture of the spermatogonia with Sertoli cells derived from a normal individual.

Work proceeds apace to produce a successful prosthetic vascular graft which would remain patent as a blood conduit, prevent thrombosis of the blood in low flow or adverse states and control anastomotic cellular hyperplasia with overgrowth of smooth muscle cells and matrix production. This has proved difficult to achieve, but there are encouraging attempts to engineer a composite vascular graft in which microvessel endothelial cells present in a small sample of subcutaneous abdominal wall fat, obtained by the relatively simple cosmetic liposuction technique, grow as an antithrombotic endothelial layer to line the surface of an expanded polytetrafluoroethylene tube. Expect development in this field, particularly for the replacement of arteries.

ASSOCIATION OF HLA TYPE WITH DISEASE

Linkage disequilibrium and disease susceptibility

An impressive body of data is accumulating which links specific HLAs with particular disease states in the human (table 17.1) and even more striking relationships may be uncovered as the complexity of the HLA-D region is unraveled. The relationships are influenced by **linkage disequilibrium**, a state where closely linked genes on a chromosome tend to remain associated rather than undergo genetic randomization in a given population, so that the frequency of a pair of alleles occurring together is greater than the product of the individual gene frequencies (figure 17.19a). This could result from natural selection favoring a particular haplotype or from insufficient time elapsing since the first appearance of closely located alleles to allow them to

Table 17.1. Association of HLA with disease. (Data mainly from Ryder *et al.*, see legend to figure 17.19, and Thorsby E. (1995) *The Immunologist* 3, 51.)

DISEASE	HLA ALLELE	RELATIVE RISK
a Class II associated		
Hashimoto's disease	DR11	3.2
Primary myxedema	DR17(*3)	5.7
Thyrotoxicosis (Graves')	DR17(*3)	3.7
Insulin-dependent diabetes	DQ8	14
	DQ2/8	20
	DQ6	0.2
Addison's disease (adrenal)	DR17(*3)	6.3
Goodpasture's syndrome	*DR2	13.1
Rheumatoid arthritis	DR4	5.8
Juvenile rheumatoid arthritis	DR8	8.1
Sjögren's syndrome	DR17(*3)	9.7
Chronic active hepatitis (autoimmune)	DR17(*3)	13.9
Multiple sclerosis	*DR2,*DQ6	12
Narcolepsy	DQ6	38
Dermatitis herpetiformis	DR17(*3)	56.4
Celiac disease	DQ2	250
Tuberculoid leprosy	*DR2	8.1
b Class I, HLA-B27 associated		
Ankylosing spondylitis	B27	87.4
Reiter's disease	B27	37.0
Post-salmonella arthritis	B27	29.7
Post-shigella arthritis	B27	20.7
Post-yersinia arthritis	B27	17.6
Post-gonococcal arthritis	B27	14.0
Uveitis	B27	14.6
Amyloidosis in rheumatoid arthritis	B27	8.2
c Other class I associations		
Subacute thyroiditis	B35	13.7
Psoriasis vulgaris	Cw6	13.3
Idiopathic hemochromatosis	A3	8.2
Myasthenia gravis	B8	4.4

*DR specificities relate to 'old nomenclature', figure 17.9.

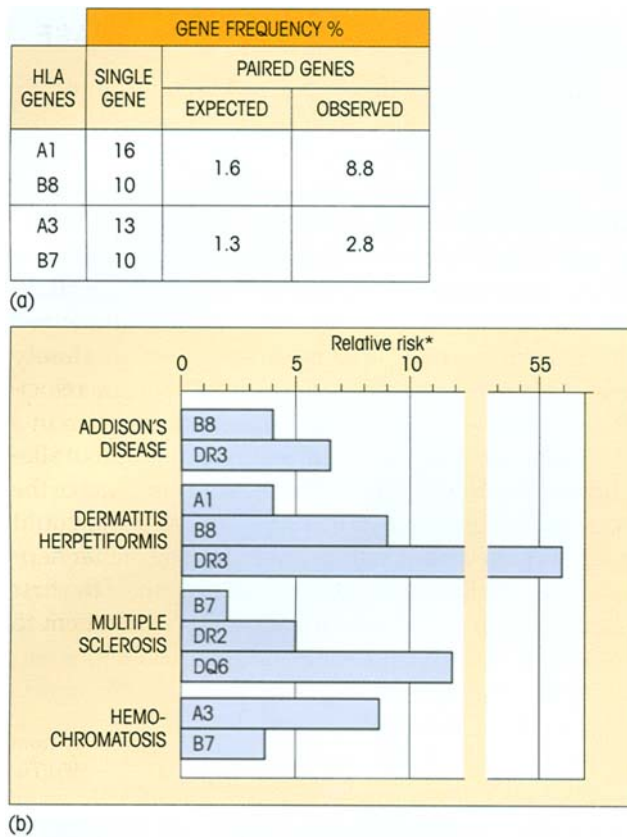


Figure 17.19. Linkage disequilibrium and the association between HLA and disease. (a) Two examples of linkage disequilibrium. The expected frequency for a pair of genes is the product of each individual gene frequency. B8 and DR3 (old nomenclature; figure 17.9) are in linkage disequilibrium as are B7 and DR2 (old nomenclature); thus the haplotypes A1, B8, DR3 and A3, B7, DR2 are relatively common. (b) Influence of linkage disequilibrium on disease association. *Relative risk = the increased chance of contracting the disease for individuals bearing the antigen relative to those lacking it. (Data from Ryder L.P., Andersen E. & Svejgaard A. (1979) HLA and Disease Registry 1979. *Tissue Antigens* supplement.)

become randomly distributed throughout the population. Be that as it may, a significant association between a disease and a given HLA specificity does not imply that we have identified the disease susceptibility gene, because we might find an even better correlation with another HLA gene in linkage disequilibrium with the first. To take an example: in multiple sclerosis, an association with the B7 allele was first established but, when patients were typed for the D locus, a much stronger correlation with DR2 emerged (figure 17.19b). The initial correlation with B7 resulted from linkage disequilibrium between B7 and DR2. We still cannot be sure that DR2 itself is the disease susceptibility gene since, carrying the argument a stage further, we cannot exclude the possibility of finding an even greater association with another closely linked gene, and indeed recent studies reveal an even tighter relationship to

DQB1*0602. However, in those instances where two susceptibility genes encoding α and β chains synergize and are in the *trans* configuration (i.e. on different chromosomes), linkage disequilibrium is less likely to be the explanation.

Ethnic studies may help by making available recombination events which alter haplotypes and permit the identification of susceptibility determinants outside their normal context. Thus, the DQ α (DQA1*0201) linked to DR7 on the Caucasian haplotype has a neutral effect on diabetes susceptibility but, in the black population, DR7 is associated with a high-risk DQ α (DQA1*0301) and now becomes a susceptibility haplotype.

Association with immunological diseases

With odd exceptions such as idiopathic hemochromatosis and congenital adrenal hyperplasia resulting from a 21-hydroxylase deficiency, which only gain membership of this august club through linkage disequilibrium, HLA-linked diseases are intimately bound up with immunological processes. By and large, the HLA-D-related disorders are autoimmune with a tendency for DR3 or linked genes to be associated with organ-specific diseases. It has been suggested that HLAs might affect the susceptibility of a cell to viral attachment or infection, thereby influencing the development of autoimmunity to associated surface components. Inevitably though, because class II genes tend to dominate these relationships, the temptation is to think in terms of immune response genes controlling the nature of the reaction to the relevant autoantigen or to whatever might be a causative agent, perhaps through an ability to bind certain antigenic peptides. A given HLA-D allele may permit the binding of foreign peptides and cross-reacting self-epitopes, or may itself contribute the self-epitope since *processed* MHC molecules appear abundantly on the cell surface. The modern MHC-peptide tetramer technology (cf. p. 138) or the alternative peptide/MHC-Ig dimer reagents should be useful for the detection of sensitized T-cells in these patients and for tracking down the identity of the relevant epitopes; diagnostic applications would follow.

Insulin-dependent diabetes mellitus is associated with DQ8 (DQA1*0301, DQB1*0302) and DQ2 (DQA1*0501, DQB1*0201), but the strongest susceptibility is seen in the **DQ2/8 heterozygote** arguing in favor of a role for the *trans*-encoded DQ molecule(s) in these individuals, most probably DQ(α 1*0301, β 1*0201). The fact that two genes are necessary to determine the strongest susceptibility, and the universal-

ity of this finding in all populations studied, implies that the DQ molecules themselves, not other molecules in linkage disequilibrium, are primarily involved in disease susceptibility. Possession of some subtypes of DQ6 gives a dominantly protective effect (table 17.1; relative risk 0.2). Thus the degree of HLA-associated predisposition to type 1 diabetes will be a composite of the effects of the particular combination of *cis*- or *trans*-encoded DQ molecules. To get down to the molecular level, different polymorphic amino acids at residues DQ α 52 and DQ β 57 have a powerful influence on disease susceptibility.

DR4 and, to a lesser extent, DR1 are risk factors for **rheumatoid arthritis** in white Caucasians. Analysis of DR4 subgroups, and of other ethnic populations where the influence of DR4 is minimal, has identified a particular linear T-cell sequence from residues 67–74 as the disease susceptibility element, and the variations observed are based on sharing of this sequence with other HLA-DR specificities, which presumably arose from the recombination and gene conversion events responsible for MHC polymorphism (table 17.2). This stretch of amino acids is highly polymorphic and forms pockets 4 and 7 of the peptide-binding cleft. There are striking differences between the sets of peptides bound by DR subtypes associated with rheumatoid arthritis and those that are not; notably, differences involve the amino acid charge at residues 4 or 5 of the peptide sequence bound to the cleft. Analysis of the peptides bound spontaneously to the DR molecules from patients could yield an important clue to pathogenesis. As in diabetes, the DR2 allele is under-represented and DR2-positive patients have less severe disease, implying that a DR2-linked gene might

be protective in some way (through interaction with a heat-shock protein maybe?).

The association with HLA in **ankylosing spondylitis** is quite remarkable; up to 95% of patients are of B27 phenotype as compared with around 5% in controls. The incidence of B27 is also markedly raised in other conditions when accompanied by sacroiliitis, e.g. Reiter's disease, acute anterior uveitis, psoriasis and other forms of infective sacroiliitis, such as *Yersinia*, gonococcal and *Salmonella* arthritis. The extraordinarily close association with B27 and the development of a somewhat similar disease in rodents made transgenic for the B27 gene speak for the B27 molecule itself as a central pathogenic factor. There are nine subtypes of B27 encoded by the B*2701–2708 alleles, all of which are associated with ankylosing spondylitis and have a 'B' pocket in their cleft which is very exclusive with respect to binding the arginine side-chain. The hunt should be on for a nonapeptide with an arginine at residue 2. The involvement of infective agents may provide a clue. The cross-reaction of B27 with *Klebsiella pneumoniae* nitrogenase reported by Ebringer and colleagues is certainly provocative in this respect. One suggestion is that a bacterial peptide may cross-react with a B27-derived sequence and provoke an autoreactive T-cell response.

Deficiencies in C4 and C2, which are MHC class III molecules, clearly predispose to the development of **immune complex disease** (see p. 308), and so it would be expected that the inheritance of null genes or alleles coding for the less active complement allotypes would increase the risk of rheumatological disorders and add yet further complexity to the correlations between HLA types and disease.

Table 17.2. Shared sequences on the DR β 1 α -helix in different haplotypes confer susceptibility to rheumatoid arthritis. (Data from Bell J.I. & McMichael A.J. (1993) In Lachmann P.J., Peters D.K., Rosen F.S. & Walport M.J. (eds) *Clinical Aspects of Immunology*, p. 748. Blackwell Scientific Publications, Oxford.)

	HLA-DR SPECIFICITY	DR β 1 GENE	HLA-DR β RESIDUES IN THE THIRD ALLELIC HYPERVARIABLE REGION									
			66	70				74				
Susceptible to RA	DR1	*0101	Asp	Leu	Leu	Glu	Gln	Arg	Arg	Ala	Ala	Val
	DR4	*0404	—	—	—	—	—	—	—	—	—	—
	DR4	*0405	—	—	—	—	—	—	—	—	—	—
	DR4	*0401	—	—	—	—	—	Lys	—	—	—	—
	DR10	*1001	—	—	—	—	Arg	—	—	—	—	—
Not susceptible to RA	DR4	*0402	—	Ile	—	—	Asp	Glu	—	—	—	—
	DR4	*0407	—	—	—	—	—	—	—	Glu	—	—
	DR7	*0701	—	Ile	—	—	Asp	—	—	Gly	Gln	—
Nucleotide sequences	DR1	*0101	GAC	CTC	CTG	GAG	CAG	AGG	CGG	GCC	GCG	GTG
	DR4	*0401	---	---	---	---	---	-A-	---	---	---	---

Last, it is worthy of note that the relationship of MHC to disease resistance and vaccine efficacy in farm animals is beginning to preoccupy veterinary scientists; it is known, for instance, that susceptibility to Marek's disease in White Leghorn chickens is associated with distinct MHC haplotypes.

REPRODUCTIVE IMMUNOLOGY

The fetus is a potential allograft

A further consequence of polymorphism in an outbred population is that mother and fetus will almost certainly have different MHCs. Some examples of selection for heterozygotes (where maternally and paternally derived haplotypes are different) over homozygotes (both fetal haplotypes identical with the mother's) in viviparous animals suggest that this is beneficial. Likewise, the placentas of F1 offspring are larger than normal when mothers are preimmunized to the paternal H2 haplotype and smaller when mothers are tolerant to these antigens.

The threat posed to the fetus as a potential graft due to the possession of paternal transplantation antigens so intrigued Lewis Thomas that he was moved to suggest that rejection of the fetus might initiate parturition, although it would be difficult to account for the normal birth of female offspring to pure-strain mating pairs where fetus and mother would have identical histocompatibility antigens without further postulating a placenta-specific surface antigen.

Nonetheless, in the human hemochorial placenta, maternal blood with immunocompetent lymphocytes does circulate in contact with the fetal trophoblast and we have to explain how the fetus avoids allograft rejection, despite the development of an immunological response in a proportion of mothers as evidenced by the appearance of anti-HLA antibodies and cytotoxic lymphocytes. In fact, prior sensitization with a skin graft fails to affect a pregnancy, showing that trophoblast cells are immunologically protected; indeed, they are resistant to most cytotoxic mechanisms although susceptible to IL-2-activated NK cells. Some of the many speculations which have been aired on this subject are summarized in figure 17.20.

Undoubtedly, the most important factor is the well-documented lack of both conventional class I and class II MHC antigens on the placental villous trophoblast which protects the fetus from allogeneic attack. These fundamental changes in the regulation of MHC genes also lead to the unique expression of the nonclassical HLA-G protein on the extravillous cytotrophoblast. This may protect the trophoblast from killing by uter-

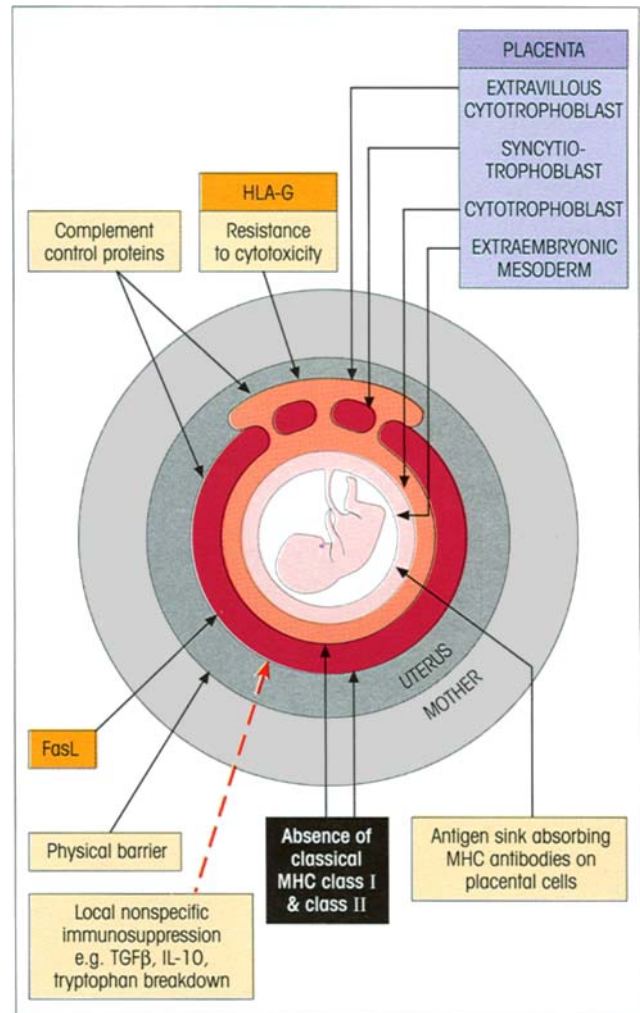


Figure 17.20. Mechanisms postulated to account for the survival of the fetus as an allograft in the mother.

ine endometrial large granular lymphocytes which are an NK cell subset, since HLA-null lymphoblastoid cells transfected with HLA-G are resistant to NK lysis. This would be consistent with the 'missing self hypothesis' which postulates that MHC class I inhibits a positive signal from a potential target to the NK cell. On the other hand, the placenta is vulnerable to IL-2-activated uterine NK cells, which may therefore fine tune overaggressive invasion by the trophoblast early in pregnancy and may control the normal development of the blastocyst in the decidualized uterus, in contrast with the excessive trophoblast proliferation and tissue destruction produced by transfer of blastocysts to mouse kidney.

Maternal IgG antipaternal MHC is found in 20% of first pregnancies and this figure rises to 75–80% in multiparous women. Some of these antibodies cross-react with HLA-G, but the vulnerability of the trophoblast cells to complement is blocked by the presence on their

surface of the control proteins which inactivate C3 convertase (cf. p. 307).

Cytokines seem to have a complex role in post-implantation pregnancy given the production of growth factors such as CSF-1 and GM-CSF, which have a trophic influence on the placenta, and of transforming growth factor- β (TGF β), which could help to damp down any activation of NK cells by potentially abortive events such as intrauterine exposure to lipopolysaccharide (LPS) or to interferons. The placenta itself produces IL-10, the importance of which was revealed by IL-10 'knockout' experiments (cf. p. 142) which caused runting of the fetus compared with its littermate controls—incidentally this is a slick demonstration that IL-10 is synthesized by the embryo rather than the trophoblast.

Other factors which may protect the fetus are the presence of Fas-ligand at the trophoblast maternal-fetal interface, and the suppression of T-cell activity through tryptophan degradation brought about by the catabolic enzyme indoleamine 2,3-dioxygenase present in trophoblast cells and macrophages.

Fertility can be controlled immunologically

Notwithstanding the complacent attitudes of some members of the chattering classes, informed opinion recognizes the enormity of the problems posed by the world population explosion. That being so, a rich variety of strategic approaches are being followed, one of which involves the targeting of key hormones (figure 17.21a) by the production of neutralizing autoantibodies.

Immunization, actually autoimmunization, with diphtheria toxoid-linked **luteinizing hormone releasing hormone** (LHRH), which regulates gonadotropin synthesis, impairs fertility in male animals and causes a marked atrophy of the prostate accompanied by a devastating fall in testosterone levels—immunological castration no less. The therapy may have a role in androgen-dependent prostatic carcinoma, while in postpartum women it should prevent ovulation and prolong lactational amenorrhea without adverse effects. Internal image anti-idiotypic Ab_{2 β} (cf. p. 208) has been raised against monoclonal antiprogesterone and shown to provoke **progesterone neutralizing antibodies** in experimental animals.

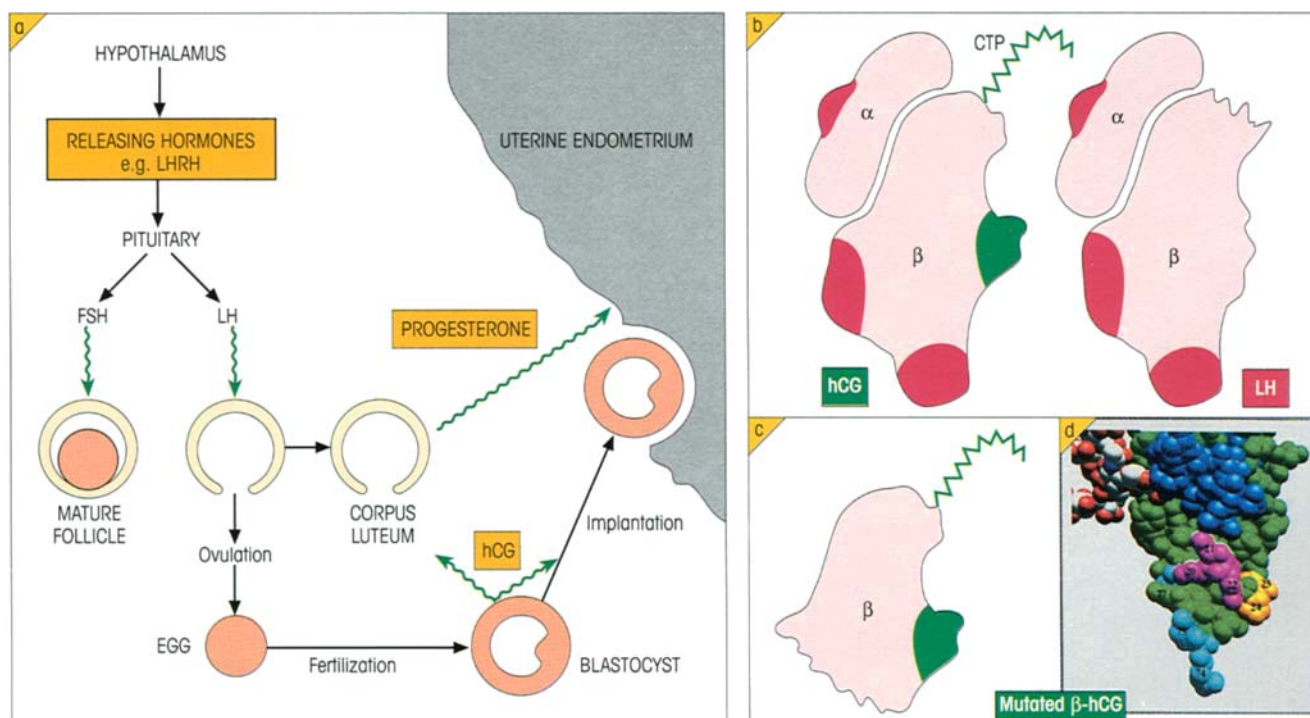


Figure 17.21. Immunological contraception. (a) Target hormones. (b) Purely diagrammatic representation of human chorionic gonadotropin (hCG) showing epitopes specific for hCG (green) and structures similar if not identical to luteinizing hormone (LH) (dark red). (c) β -hCG mutant in which the hCG-specific epitopes have been preserved but epitopes shared with LH lost. As a candidate vaccine

this should avoid the problem of inducing antibodies cross-reacting with LH. (d) Structure of β -hCG showing some of the residues altered (magenta, yellow and cyan) to produce epitope-loss mutants, fulfilling the criteria in (c). (Reproduced from Jackson A.M. *et al.* (1996) *Journal of Reproductive Immunology* 31, 21, with permission from Elsevier Science, Ireland Ltd.)

Most attention has focused on a human vaccine based on **human chorionic gonadotropin** (hCG) which is made by the preimplantation blastocyst and is essential for the establishment of early pregnancy (figure 17.21a); it is also the antigen assayed in the urine tests for pregnancy which cause so much pleasure or consternation as the case may be. The α chain of hCG is common to follicle-stimulating hormone (FSH), thyroid-stimulating hormone (TSH) and LH while the β -subunit shows around 80% homology with LH (figure 17.21b). Clearly immunization with whole hCG would elicit some exceedingly unwelcome reactions and two vaccines try to avoid them wholly or in part. Both couple the antigen to tetanus or diphtheria toxoid carriers. The WHO vaccine uses the 37-amino acid C-terminal peptide (CTP) which is wholly specific for hCG but not

exciting as an immunogen. The Indian vaccine combines β -hCG with ovine α chain. This evokes much better antibody responses, and in Phase 2 clinical trials only one pregnancy was observed in 1224 cycles when neutralizing antibodies were above a certain level. Despite strong cross-reactions with LH there was said to be no disturbance of ovulation or loss of libido. As contraceptive cover while antibodies were rising, cell-mediated hypersensitivity was induced by local intrauterine implantation of purified extracts from Neem, an ancient Indian tree. As antibody levels fall, fertility is regained. Long-term maintenance of adequate antibody titers should be possible with biodegradable microspheres or a recombinant *Salmonella* construct incorporating the β -hCG gene.

SUMMARY

Graft rejection is an immunological reaction

- It shows specificity, the second set response is brisk, it is mediated by lymphocytes and antibodies specific for the graft are formed.

Genetic control of transplantation antigens

- In each vertebrate species there is a major histocompatibility complex (MHC) which is responsible for provoking the most intense graft reactions.
- Parental MHC antigens are codominantly expressed on cell surfaces.

Other consequences of MHC incompatibility

- Class II MHC molecules provoke a mixed lymphocyte reaction of proliferation and blast transformation when genetically dissimilar lymphocytes interact.
- Class II differences are largely responsible for the reaction of tolerated grafted lymphocytes against host antigen (graft-vs-host (g.v.h.) reaction).
- Siblings have a 1:4 chance of identity with respect to MHC.

Mechanisms of graft rejection

- Preformed antibodies cause hyperacute rejection within minutes.
- CD8 lymphocytes play a major role in the acute early rejection of first set responses.
- The strength of allograft rejection is due to the surprisingly large number of allospecific precursor cells. These derive mainly from the variety of T-cells which recognize

allo-MHC plus self-peptides plus a small number which directly recognize the allo-MHC molecule itself (the direct pathway); later rejection increasingly involves allogeneic peptides presented by self-MHC (the indirect pathway).

- Acute late rejection of organ grafts from 11 days onwards is caused by Ig and C binding to graft vessels.
- Insidious and late rejection may be due to immune complex deposition or breakthrough against immunosuppression.

Prevention of graft rejection

- This can be minimized by cross-matching donor and graft for ABO and MHC tissue types. Individual MHC antigens are typed by the PCR using discriminating primer pairs.
- Rejection can be blocked by agents producing general immunosuppression such as antimetabolic drugs (e.g. azathioprine), anti-inflammatory steroids and antilymphocyte monoclonals. Cyclosporin A, FK506 and rapamycin represent exciting groups of T-cell-specific drugs; complexes of cyclosporin and FK506, with their cellular ligands, block calcineurin, a phosphatase which activates the IL-2 transcription factor NFAT, while rapamycin complexes with the FK-binding protein to block kinases involved in cell proliferation.
- Antigen-specific depression through tolerance induction can be achieved with injection of allogeneic bone marrow with costimulatory blockade by anti-CD154 (CD40L) plus a CTLA-4-Ig fusion protein. Dendritic cell precursors

can also induce tolerance through antigen presentation in the absence of B7 costimulators.

Tissue engineering

- A distinct goal is to create autografts from stem cells or embryonically reprogramed adult nuclei differentiating in culture under the influence of specified growth factors.

Xenografting

- Strategies are being developed to prevent hyperacute rejection of pig grafts in humans due to reaction of natural antibodies in the host with galactose epitopes on pig cells and acute vascular rejection by acquired antibodies produced by the xenogeneic antibody response.

Clinical experience in grafting

- Cornea and cartilage grafts are avascular, produce local immunosuppressive factors and are comparatively well tolerated.
- Kidney grafting gives excellent results and has been the most widespread, although immunosuppression must normally be continuous.
- High success rates are also being achieved with heart and liver transplants particularly helped by the use of cyclosporin. Lung and pancreatic β -cell grafts are less successful.
- Bone marrow grafts for immunodeficiency and aplastic anemia are accepted from matched siblings, but it is difficult to avoid g.v.h. disease with allogeneic marrow without first purging CD8 T-cells in the graft or preferably by inducing tolerance using costimulatory blockade. Recombinant G-CSF and GM-CSF accelerate the return of myeloid cells after grafting.
- Experimental studies on transplantation of neural tissue and the production of prosthetic vascular grafts have been described.

Association of HLA type with disease

- HLA specificities are often associated with particular diseases, e.g. HLA-B27 with ankylosing spondylitis, B8

with myasthenia gravis, DR17 with Sjögren's syndrome, DR4 with rheumatoid arthritis, DQ2 and 8 with insulin-dependent diabetes mellitus and DR2, DQ6 with multiple sclerosis.

- The association may be related to an ability to bind particular antigenic peptides or to cross-react with certain infectious agents.

The fetus as an allograft

- Differences between MHC of mother and fetus may be beneficial to the fetus but as a potential graft it must be protected against transplantation attack by the mother.
- A major defense mechanism is the lack of classical class I and II MHC antigens on syncytiotrophoblast and cytotrophoblast which form the outer layers of the placenta.
- The extravillous cytotrophoblast expresses a nonclassical nonpolymorphic MHC class I protein, HLA-G, which may act to inhibit cytotoxicity by maternal NK cells.
- Syncytio- and cytotrophoblasts bear surface complement regulatory proteins which break down C3 convertase and so block any complement-mediated damage.
- Local production of IL-10 and TGF β , tryptophan degradation and the presence of FasL may suppress unwanted reactions.

Immunological contraception

- Autoimmunization against hormones such as LHRH and hCG, which play a key role in reproduction, can block fertility.
- The C-terminal peptide of β -hCG and a complex of human β -hCG with sheep α chain have been used in human trials. If antibody titers are sufficiently high, prevention of pregnancy is close to 100% in the latter case. As antibody levels fall, fertility is regained.

See the accompanying website (www.roitf.com) for multiple choice questions.

FURTHER READING

- Austen K.F., Burakoff S.J., Rosen F.S. & Strom T.B. (eds) (1996) *Therapeutic Immunology*. Blackwell Science, Oxford.
- Bodmer J.G., Marsh S.G., Albert E.D. *et al.* (1994) Nomenclature for factors of the HLA system. *Tissue Antigens* **44**, 1.
- Borel J.F. *et al.* (1996) *In vivo* pharmacological effects of cyclosporin and some analogues. *Advances in Pharmacology* **35**, 115. (An in-depth review of the field by the discoverer of cyclosporin and his colleagues.)

- Daar A.S. (Chairman) (1999) Animal-to-human organ transplants—a solution or a new problem? (An in-depth round table discussion.) *Bulletin of the World Health Organization* **77**, 54.
- Friend P.J. *et al.* (1999) Phase I study of an engineered aglycosylated humanized CD3 antibody in renal transplantation rejection. *Transplantation* **68**, 1632.
- Gurdon J.B. & Colman A. (1999) The future of cloning. *Nature* **402**, 743.
- Kahn A. (2000) Converting hepatocytes to β -cells—a new approach for diabetes? *Nature Medicine* **6**, 505.